TestBoard_ADC124
PCB behaviour tests

Tests acquired 15 October 2026. Explanatory report revision: 15 October 2026.

Connected PZT strips, left untouched. Temporary diagnostic firmware; LPSPI only. The original matrix is retained; Tests 21-23 add new long-VMID recordings from the same setup.

Completed work	Result
Original test matrix	256 programmed conditions / 762 complete recordings
Added VMID tests	Tests 21-23: 63 additional recordings / 21 programmed conditions; combined 825 recordings.
Original acquisition	7,919,928 cycles acquired; 7,919,811 received; 117 counted whole-cycle USB discards
Integrity	Zero transfer errors, data errors, USB-write errors and controlled timing overruns
Board afterwards	Original production firmware and settings restored; acquisition stopped

What this report is trying to explain

Why baseline readings differ between arrays and MUXes; how external switching, sensor connection time, ADC read order and conversion activity change those readings; and whether quiet recordings contain bursts or isolated spikes.

Main findings

External MUX address 6 provides an unintentional reference visit on MUXes 1-3. Holding an address gives near-midscale baselines. Array B loses much more baseline after a long disconnected interval. Repeated conversions shift readings even with VMID held continuously. Isolated spikes remain in the VMID-reset recording. The tests identify operating-history effects; they do not identify a single defective component.

Each numbered test starts with its purpose, exact procedure, parameters, measurement and legend. The following results page gives the graphs or tables and explains what can, and cannot, be concluded.

Added Tests 21-23 cover a 20-second timed VMID hold, ten consecutive conversions per selected MUX path, and regular full scans with 1, 5 and 10 ms VMID waits.

Contents

Terminology and the electrical path . . . 3
Board setup and the special wiring of address 6 . . . 4
Timing, statistics and graph conventions . . . 5
01. Full scans with and without external MUX address 6 . . . 6
02. Baseline spread between ADC inputs at each external MUX address . . . 8
03. Boundary controls with equal phase slots . . . 10
04. Fixed-address acquisition: external switching removed . . . 13
05. First eight conversions after a VMID connection . . . 16
06. External MUX address order with equal phase slots . . . 22
07. Fast external MUX address order and placement of the cycle gap . . . 24
08. ADC array order and ADC input order . . . 26
09. One active array versus both arrays . . . 28
10. VMID connection versus extra conversions: 50 µs interval . . . 30
11. VMID connection versus extra conversions: 20 µs interval . . . 32
12. MUX isolation and reconnection at an unchanged address . . . 34
13. Known-voltage comparison of external MUX addresses 6 and 7 . . . 37
14. Repeated conversions while continuously connected to VMID . . . 39
15. Long recordings: time behaviour and sustained disturbances . . . 42
16. Long recordings: isolated spikes and maximum excursions . . . 44
17. Long recordings: power spectral density . . . 47
18. Long recordings: raw versus quiet noise statistics . . . 49
19. Correlation between measured paths . . . 52
20. Timing stability and association with code changes . . . 54
21. One full cycle after a 20-second timed VMID hold . . . 56
22. Ten conversions of one fixed ADC input after the long hold . . . 59
23. Regular full cycles with 1, 5 and 10 ms VMID settling . . . 63
Original matrix: acquisition checks, restoration and evidence files . . . 70
How to find the underlying data . . . 71
Device references and limits of component attribution . . . 72
Appendix: original programmed conditions and measured cycle periods . . . 73

Terminology and the electrical path

Sensor connection to external MUX input at an address to MUX common output to OPA365 buffer to ADC input.

Term	Meaning used throughout this report
Array A / Array B	Each array has four external MUXes, four buffers and one ADC. ADC A measures Array A; ADC B measures Array B.
MUX number 1, 2, 3 or 4	Which external MUX chip within an array. Its buffered output is wired to the same-numbered ADC input: MUX 1 to ADC IN1, ..., MUX 4 to ADC IN4.
External MUX address 0-7	Which input pin is connected to the common output of each external MUX. Address numbering is zero-based. Address 5 means the sixth input pin; it does not mean ADC IN4 or MUX 5.
ADC input IN1-IN4	Which buffered MUX output the ADC converts. Changing ADC input does not change the external MUX address.
VMID	Nominal 1.65 V reference/bias. Selecting external MUX address 7 connects VMID through each external MUX.
MUX EN disabled	External MUX signal paths are open. Buffers and ADCs remain powered; the MUX-output bias resistors remain connected to VMID. This is not ADC chip-select or ADC power-off.
ADC A first / ADC B first	At one external MUX address, which ADC is read first on the shared SPI bus. The other ADC is then read.

Within an array, all four external MUXes share address/enable control. In these two-array tests, the arrays normally change address together. The SPI reads are sequential, not simultaneous.

Board setup and the special wiring of address 6

Item	Array A	Array B
External MUX type	TMUX1108	TMUX1308A
Bias resistor: MUX 1 / 2 / 3 / 4	470 kohm / 1 Mohm / 470 kohm / 1 Mohm	1 Mohm / 470 kohm / 1 Mohm / 470 kohm
Buffer / ADC	OPA365 / ADC124S101	OPA365 / ADC124S101
External MUX addresses 0-5	Sensor inputs on MUXes 1-4	Sensor inputs on MUXes 1-4
External MUX address 6	VMID on MUXes 1-3; sensor on MUX 4	VMID on MUXes 1-3; sensor on MUX 4
External MUX address 7	VMID on every MUX	VMID on every MUX

On MUXes 1-3, the address-6 input is physically tied to the address-7 VMID input. Consequently, a full address-0-to-6 scan still connects those MUXes to VMID once per cycle even when software says VMID off. MUX 4 instead has a seventh sensor at address 6. There are 25 sensors per array, 50 total.

Both arrays use the 3.3 V TPS7A2033PDBVR LDO. REF2033 supplies nominal 3.3 V ADC reference/supply and 1.65 V VMID. The user reports 100 nF at the REF2033 VMID pad, and 150 pF to ground plus a nearby 100 ohm isolation resistor at each MUX VMID input. The precise resistor/capacitor net ordering has not been verified from a schematic; no inferred schematic is used here.

The PZT strips remained attached and untouched. Table vibration was not monitored. MUX type is tied to array location, ADC, buffers and layout; an array difference is not proof that the MUX type alone caused it.

Timing, statistics and graph conventions

Term / symbol	Exact interpretation
Phase slot	A fixed interval beginning before external MUX switching. It includes switching guards, the configured wait and any ADC reads. A 100 µs slot is not 100 µs added after the reads.
Cycle period	Start-to-start interval for repeating the full programmed sequence. Any remaining time includes encoding/enqueue and waiting; the last analog connection persists unless explicitly disabled.
Settling delay d	The programmed wait after a switching request and before ADC reads. The wait still executes if the address did not change. ADC B normally starts later than ADC A because they share SPI.
Baseline	For each route and recording: median ADC code. For repeated conditions: median of the three recording medians.
MUX baseline center	Median of six separately measured sensor baselines at external MUX addresses 0-5, for one MUX. It is not a temporal SD.
Baseline spread	Maximum baseline minus minimum baseline across the specified inputs or external MUX addresses. This is spatial variation, not temporal noise.
SD / peak-to-peak	Temporal sample standard deviation (ddof=1) / maximum minus minimum recorded code. Repeated-condition tables use the median of per-record statistics.
Units	1 count = 3.3 / 4096 V = 0.805664 mV nominally. Midscale 2048 counts corresponds to nominal VMID. mV values are code conversions, not separate voltmeter measurements.
Lines and circles	Unless stated otherwise, circles are measured summary points. Lines connect them for readability; they are not additional measurements or time waveforms.

SPI frequencies are requested settings; SCLK edges were not measured. Timing markers bracket driver operations, not exact analog sample apertures. Diagnostic logging adds overhead, so these rates are not production sampling-rate benchmarks.

01. Full scans with and without external MUX address 6

Purpose

Check whether the hidden VMID visit at external MUX address 6 explains why MUX 4 behaves differently from MUXes 1-3.

What was done

1. Run ascending external MUX addresses 0-6 with software VMID off. Read ADC IN1-IN4 at addresses 0-5; read only ADC IN4 at address 6.

2. Repeat using only external MUX addresses 0-5, with software VMID off.

3. Repeat the full 0-6 sequence, adding external MUX address 7 and four discarded conversions per ADC after every address.

Test parameters

Parameter	Setting
Timing	No fixed phase slots or cycle period: run freely. Settling delays 1, 5 and 50 µs; graph below shows 5 µs.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

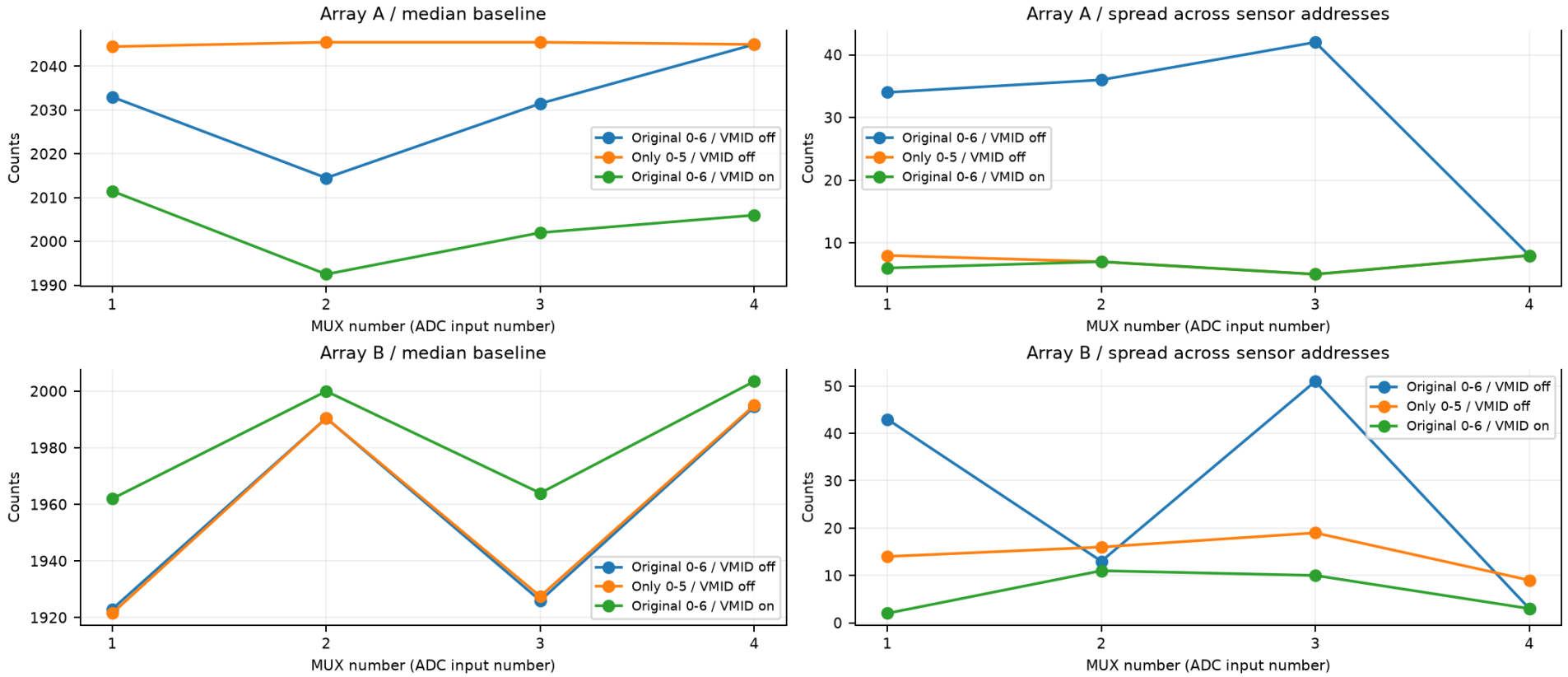
What was measured

For each MUX, compare the baseline center and max-minus-min baseline spread across the same external MUX addresses 0-5 in all conditions.

How to read the results

X = MUX number, also its ADC input number. Left = baseline center; right = spread across addresses 0-5. Upper row = Array A; lower = Array B. Each line is a complete acquisition condition, not a separate sensor.

01. Full scans with and without address 6 - results



Legend and axes

Blue = addresses 0-6, software VMID off. Orange = addresses 0-5, software VMID off. Green = addresses 0-6 with explicit VMID visits between addresses.

Conclusions from these results

On Array A / MUX 2, removing address 6 reduces spread from 36 to 7 counts; MUX 4 stays at 8 counts. Hidden reference visits explain much of the MUX-4 distinction.

Array B retains a large baseline deficit without address 6. The hidden visit does not explain its entire deficit. Removing address 6 also changes cycle time and sensor connection duty; Test 3 controls the phase durations. It loses both arrays' seventh MUX-4 sensors, so this is not a full-50-sensor production fix.

02. Baseline spread between ADC inputs at each external MUX address

Purpose

Determine how differently the four buffered MUX outputs read when their external MUXes select the same address.

What was done

1. Use the same three acquisition conditions as Test 1, separately at settling delays 1, 5 and 50 µs.

2. At each external MUX address 0-5, obtain a baseline for ADC IN1, IN2, IN3 and IN4 in the same array.

3. Subtract the smallest of those four baselines from the largest.

Test parameters

Parameter	Setting
Timing	Same freely running recordings as Test 1; this section is another analysis of those data, not another acquisition.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

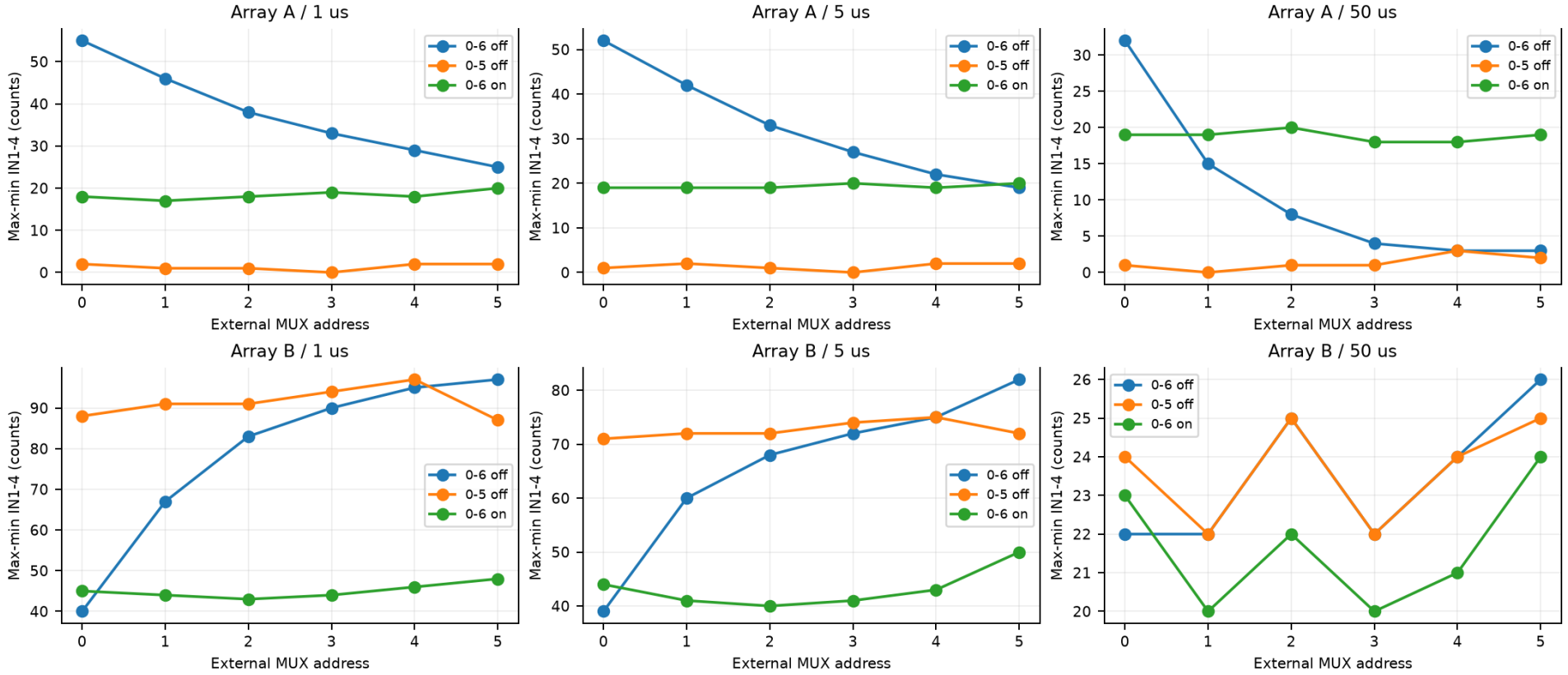
What was measured

One max-minus-min baseline difference per array, condition and external MUX address. Individual baselines are estimated over time before taking the difference.

How to read the results

X = external MUX address, not time or ADC input. Y = spread among ADC IN1-IN4 in counts. Columns = settling delay; rows = array. The four ADC inputs are separate physical paths, each connected to a different sensor at that address.

02. Baseline spread at each external MUX address - results



Legend and axes

Blue = addresses 0-6, software VMID off; orange = only addresses 0-5, off; green = full scan with explicit VMID between addresses. Lines join different external MUX addresses.

Conclusions from these results

At short delays with addresses 0-6 and software VMID off, Array A's spread generally falls as the address advances, while Array B's generally rises. This is an address/history-dependent mismatch between paths; it is not a rising or falling noise SD.

Removing address 6 or adding explicit VMID changes these shapes. A sequence of points does not show charge accumulating in one sensor: successive addresses select different sensors, and the predecessor history also changes.

03. Boundary controls with equal phase slots

Purpose

Compare different connections between the end of one sweep and the start of the next, without changing the nominal cycle duration.

What was done

1. Select external MUX addresses 0, 1, 2, 3, 4, 5 in order. At each, switch both arrays together, wait d, then read ADC A IN1-IN4 followed by ADC B IN1-IN4.

2. Give each address a 100 µs phase slot. Add one 100 µs boundary slot in one of the five states explained on the next page.

3. Repeat the seven-slot sequence every 710 µs; leave the boundary state in place during the remaining cycle overhead/wait.

Test parameters

Parameter	Setting
Timing / delays	Seven 100 µs slots; period 710 µs. Separate runs for d = 1, 5 and 50 µs.
Recorded inputs	Only external MUX addresses 0-5, all four ADC inputs on both arrays. Boundary conversions, when present, are discarded.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

What was measured

For each MUX, the median of its six sensor baselines at external MUX addresses 0-5.

How to read the results

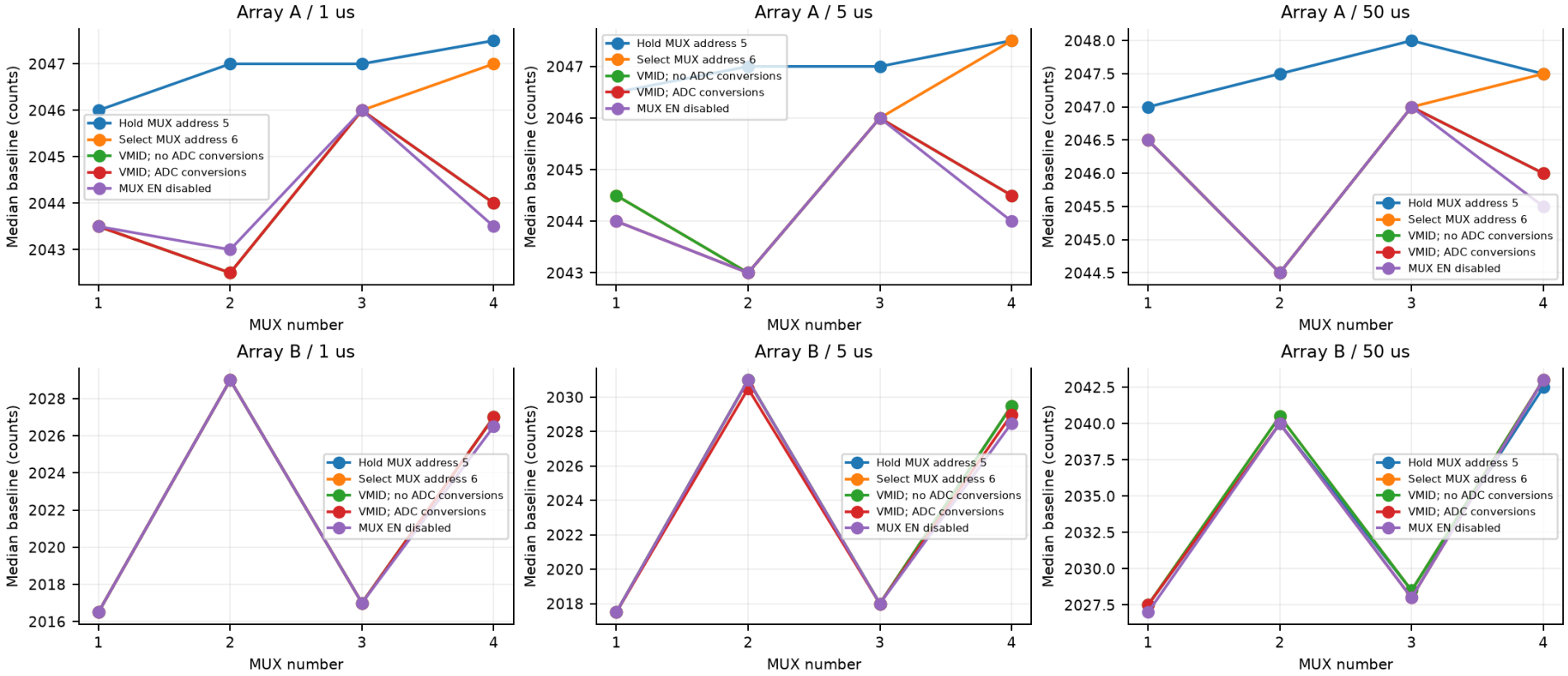
X = MUX number; Y = MUX baseline center in counts. Rows = array; columns = d. The boundary is applied once per sweep, not between every pair of addresses.

03. Boundary-state legend - what each connection means

Graph line	Exact action during the final 100 µs slot
Hold MUX address 5	Keep every MUX enabled at external MUX address 5. No ADC reads. Address 5 therefore remains connected for about 200 µs across its normal and boundary slots, plus the remaining cycle overhead/wait.
Select MUX address 6	Select address 6 on every MUX; no ADC reads. MUXes 1-3 connect VMID; MUX 4 connects its seventh sensor.
VMID; no ADC conversions	Select external MUX address 7 on every MUX. Its analog VMID connection is present, but no conversion clocks are sent during this slot.
VMID; ADC conversions	Make the same address-7 VMID connection. After wait d, read IN1-IN4 on each ADC and discard those eight results.
MUX EN disabled	Open both arrays' external MUX signal paths. No address-pin change to 7 and no ADC reads. The buffers remain biased to VMID through their resistors.

“No ADC conversions” does not mean the ADC is powered off. “VMID” here describes an analog connection through the external MUX. The equal slot durations separate that connection from the extra conversion activity. The hold control deliberately gives address 5 longer connection time.

03. Boundary controls with equal phase slots - results



Conclusions from these results

At 5 µs on Array A / MUX 2, the baseline center is 2047 counts when holding MUX address 5, versus 2043 counts with a VMID boundary. Array B's 5 µs curves nearly overlap. The boundary state has a stronger effect on A in this schedule.

Adding ADC conversions to the otherwise equal VMID boundary changes centers by at most 0.5 count across these tested delays. This applies to a 100 µs boundary, not automatically to a much shorter production VMID interval.

04. Fixed-address acquisition: external switching removed

Purpose

Check whether the large scanning offsets persist when a sensor stays connected, and whether cycling ADC inputs contributes additional offsets.

What was done

1. Select one external MUX address on all eight MUXes and leave them enabled there for the whole recording. No VMID visits or EN toggles occur during the steady recording.

2. In the four-input variant, every 710 µs read ADC A IN1-IN4, then ADC B IN1-IN4. The external address stays fixed.

3. In separate one-input runs, read only IN1 on each ADC, then only IN2, IN3 or IN4. All eight external MUXes still remain connected.

4. Repeat at external MUX addresses 0, 2 and 5. Also hold address 4 with four-input reads. Every plotted address is a separate recording, not a scan within one recording.

Test parameters

Parameter	Setting
Timing	100 µs read-phase slot; period 710 µs. The connection persists for the full period. A 5 µs wait executes before every read even without switching.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

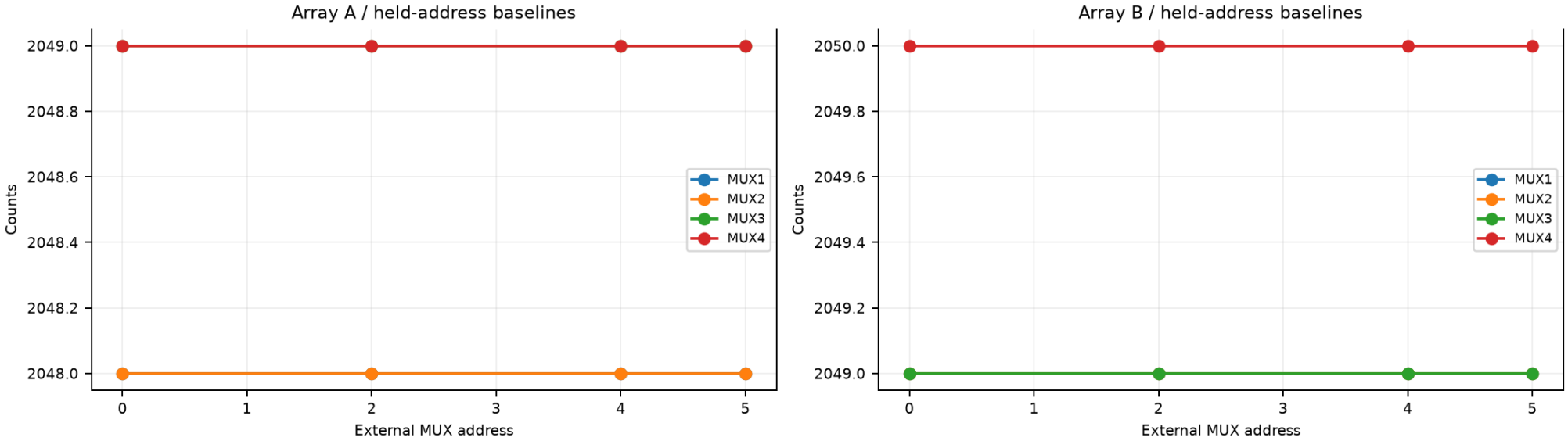
What was measured

Baseline and raw temporal SD for each selected path. The table compares the same address-2 sensor under full scanning, held/four-input and held/one-input acquisition.

How to read the results

Graph X = fixed external MUX address tested in separate runs; Y = baseline. Each colour is one MUX number. The one-input table column combines four separate selected-ADC-input recordings.

04. Fixed-address acquisition - results



Legend and axes

Each MUX line joins independent held-address recordings. Those connecting lines do not mean the firmware swept the addresses during a recording. Some MUX curves coincide exactly and cover one another; they are not missing measurements.

Conclusions from these results

All single-ADC-input, held-address-2 baselines are about 2048 counts. Four-input held baselines are about 2048-2050 counts. The large scanning deficits therefore depend on dynamic operating history.

Holding also gives the sensor continuous bias connection. Four-input versus one-input reads change both ADC input switching and conversion count (eight versus two per cycle). The differences cannot be assigned uniquely to one of those effects.

04. Same external MUX address 2 - numerical comparison

Every baseline / SD pair below is in counts and refers to the same physical address-2 sensor on that row. “Only this ADC input” means one selected input on each ADC per recording; it does not mean only one array was active.

Array / MUX / ADC input	Scanning 0-6: baseline / SD	Held address 2, four ADC inputs	Held address 2, only this ADC input
A / MUX 1 / IN1	2,030.00 / 0.027	2,048.00 / 0.000	2,048.00 / 0.000
A / MUX 2 / IN2	2,011.00 / 0.000	2,048.00 / 0.096	2,048.00 / 0.000
A / MUX 3 / IN3	2,029.00 / 0.184	2,049.00 / 0.152	2,048.00 / 0.000
A / MUX 4 / IN4	2,044.00 / 0.000	2,049.00 / 0.000	2,048.00 / 0.000
B / MUX 1 / IN1	1,925.00 / 0.414	2,049.00 / 0.169	2,048.00 / 0.029
B / MUX 2 / IN2	1,989.00 / 0.103	2,049.00 / 0.041	2,048.00 / 0.000
B / MUX 3 / IN3	1,928.00 / 0.048	2,049.00 / 0.000	2,048.00 / 0.053
B / MUX 4 / IN4	1,993.00 / 0.007	2,050.00 / 0.000	2,048.00 / 0.038

The scanning comparator is the freely running 5 µs / requested 16 MHz / software-VMID-off sequence from Test 1. Its cadence and sensor connection duty differ from the held tests. Near-zero recorded SD can occur when the ADC remains in one code; it does not establish zero analog noise.

05. First eight conversions after a VMID connection

Purpose

Separate recovery with elapsed time from recovery associated with actually performing ADC conversions after reconnecting a sensor.

What was done

1. Use one array and one ADC input per recording; disable the other array. All four MUXes in the active array still share the same address changes.

2. Select external MUX address 7, wait 100 µs, and keep that reference phase 120 µs long. Perform no conversions there.

3. Select external MUX address 2, wait d, then read the same selected ADC input eight times consecutively. Retain every result; keep this phase 100 µs long.

4. In separate controls, replace the eight reads with one read after waiting d + seven nominal word times, or that wait plus another 1 µs. Repeat this VMID-to-address-2 cycle.

Test parameters

Parameter	Setting
Timing / clock	Period 250 µs; reference slot 120 µs; sensor slot 100 µs. At 16 MHz: d = 1, 5, 50 µs. At 8 MHz: d = 1, 5 µs.
Wait-only controls	Extra wait = 7 µs at requested 16 MHz, or 14 µs at requested 8 MHz. The second endpoint adds a further 1 µs.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

What was measured

A separate median code for each of readings 1-8, across repeated cycles, then across three recordings. This is a repeatedly driven steady cycle, not a board-startup charging measurement.

How to read the results

Panels identify the array and MUX number / ADC input. Colour identifies initial wait d. Circles joined by lines are the eight-read test; x and hollow diamond are independent single-read controls. Detailed marker key follows.

05. Recovery marker key - the controls are real measurements

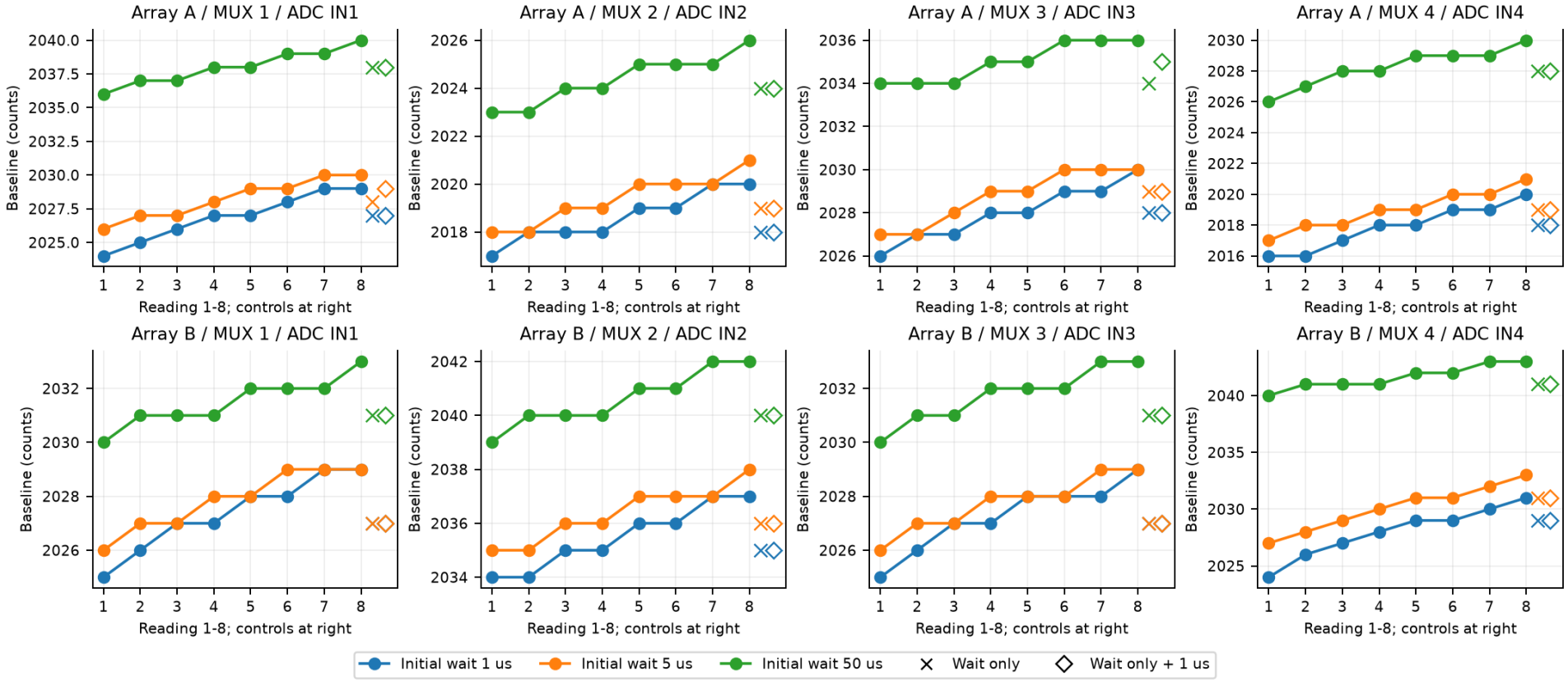
Symbol	What was acquired	Why / where plotted
Circle joined by a line	Eight consecutive conversions of one unchanged ADC input after switching from VMID to external MUX address 2.	X = actual conversion position 1 through 8. These are medians over cycles, not eight points from one selected transient.
x	One conversion after an extra seven nominal word times of waiting, with no conversions during that wait.	Compare its code with the eighth circle. Plotted just right of 8 only to keep symbols separate; it is not a ninth conversion.
Hollow diamond	One conversion after the x-control wait plus another 1 µs.	An additional wait endpoint to check sensitivity to imperfect nominal timing matching. Also plotted just right of 8 for visibility.

About the old report's apparent asterisks: the old figure used an x and a + at exactly the same horizontal position. When they overlapped, they could look like a *. They were measured wait-only controls, not error bars or statistical significance marks. This revision replaces + with a hollow diamond and separates the control positions horizontally.

Marker colour uses the same initial delay as its corresponding curve. The comparison controls are not connected to the eight-reading curve. Their horizontal offset is graphical only: it does not encode elapsed time.

These endpoints are approximately time matched using the requested word period. Driver overhead and actual sample apertures were not independently measured, so a difference from the eighth reading is evidence to investigate conversion activity, not proof that all of the difference is an ADC-specific settling error.

05. First eight conversions - requested 16 MHz - results



Legend and axes

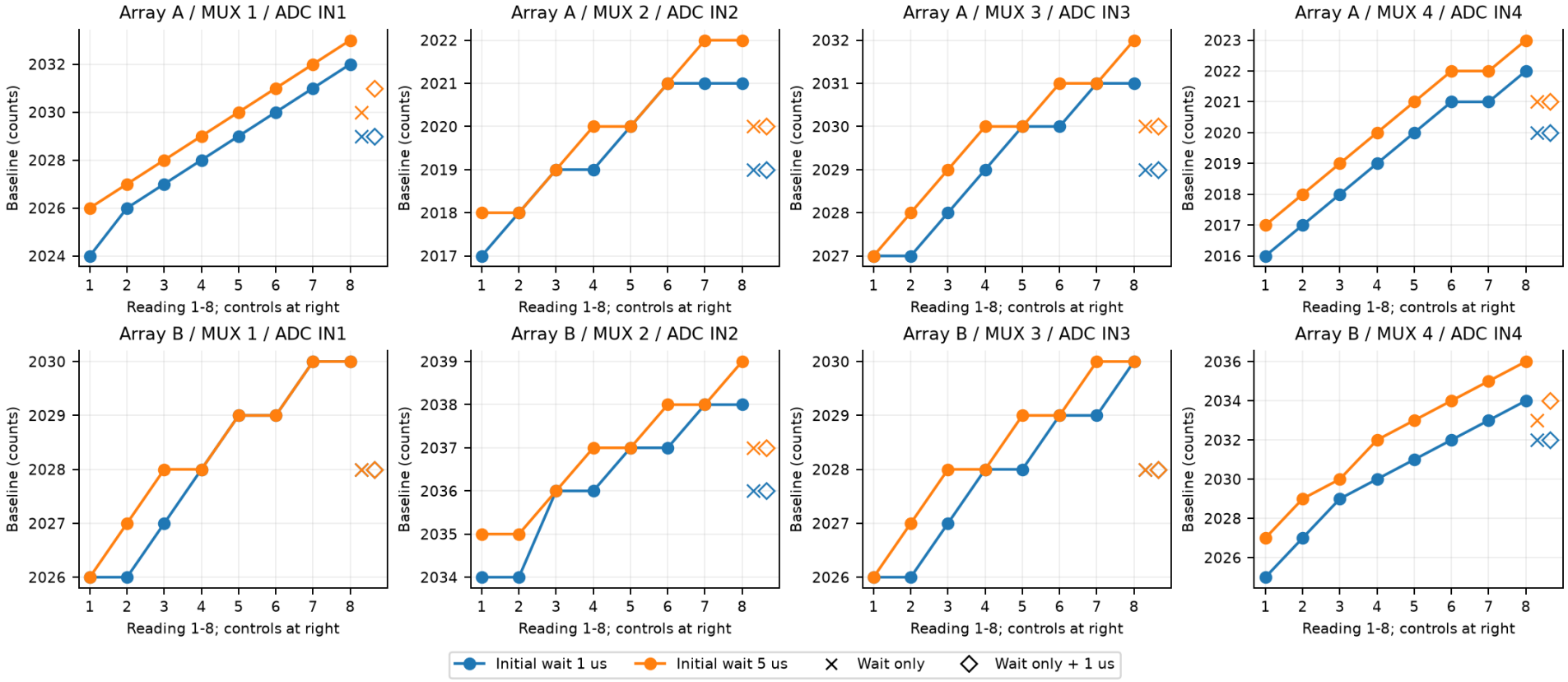
Colour = initial wait. Circles/line = eight consecutive reads. x = single read after extra waiting. Hollow diamond = single read after another 1 µs. Control markers to the right of position 8 are not extra burst readings.

Conclusions from these results

Codes generally rise through the eight-reading burst. Across both clock settings and all paths, the eighth baseline is 2-9 counts above the first (about 1.6-7.3 mV). Waiting alone does not reproduce every eighth-reading endpoint.

More conversions also change reference/supply loading and buffer activity. Test 14 shows a roughly two-count within-burst change even with VMID held continuously, so these sensor curves cannot be interpreted as sensor settling alone.

05. First eight conversions - requested 8 MHz - results



Legend and axes

Colour = initial wait. Circles/line = eight consecutive reads. x = single read after extra waiting. Hollow diamond = single read after another 1 µs. Control markers to the right of position 8 are not extra burst readings.

Conclusions from these results

Codes generally rise through the eight-reading burst. Across both clock settings and all paths, the eighth baseline is 2-9 counts above the first (about 1.6-7.3 mV). Waiting alone does not reproduce every eighth-reading endpoint.

More conversions also change reference/supply loading and buffer activity. Test 14 shows a roughly two-count within-burst change even with VMID held continuously, so these sensor curves cannot be interpreted as sensor settling alone.

05. Recovery comparison - numerical endpoints

All codes are baseline medians. “Wait” and “wait + 1 µs” are separate single-conversion recordings, as defined in the marker key. Gain = eighth reading minus first reading. Exact burst-start markers remain in channels.csv.

Array / MUX	d (µs)	Requested MHz	First	Eighth	Wait	Wait + 1 µs	Gain (counts)
A / 1	1	16	2,024.00	2,029.00	2,027.00	2,027.00	5.00
A / 1	50	16	2,036.00	2,040.00	2,038.00	2,038.00	4.00
A / 1	5	16	2,026.00	2,030.00	2,028.00	2,029.00	4.00
A / 1	1	8	2,024.00	2,032.00	2,029.00	2,029.00	8.00
A / 1	5	8	2,026.00	2,033.00	2,030.00	2,031.00	7.00
A / 2	1	16	2,017.00	2,020.00	2,018.00	2,018.00	3.00
A / 2	50	16	2,023.00	2,026.00	2,024.00	2,024.00	3.00
A / 2	5	16	2,018.00	2,021.00	2,019.00	2,019.00	3.00
A / 2	1	8	2,017.00	2,021.00	2,019.00	2,019.00	4.00
A / 2	5	8	2,018.00	2,022.00	2,020.00	2,020.00	4.00
A / 3	1	16	2,026.00	2,030.00	2,028.00	2,028.00	4.00
A / 3	50	16	2,034.00	2,036.00	2,034.00	2,035.00	2.00
A / 3	5	16	2,027.00	2,030.00	2,029.00	2,029.00	3.00
A / 3	1	8	2,027.00	2,031.00	2,029.00	2,029.00	4.00
A / 3	5	8	2,027.00	2,032.00	2,030.00	2,030.00	5.00
A / 4	1	16	2,016.00	2,020.00	2,018.00	2,018.00	4.00
A / 4	50	16	2,026.00	2,030.00	2,028.00	2,028.00	4.00
A / 4	5	16	2,017.00	2,021.00	2,019.00	2,019.00	4.00
A / 4	1	8	2,016.00	2,022.00	2,020.00	2,020.00	6.00
A / 4	5	8	2,017.00	2,023.00	2,021.00	2,021.00	6.00
B / 1	1	16	2,025.00	2,029.00	2,027.00	2,027.00	4.00
B / 1	50	16	2,030.00	2,033.00	2,031.00	2,031.00	3.00
B / 1	5	16	2,026.00	2,029.00	2,027.00	2,027.00	3.00

Array / MUX	d (µs)	Requested MHz	First	Eighth	Wait	Wait + 1 µs	Gain (counts)
B / 1	1	8	2,026.00	2,030.00	2,028.00	2,028.00	4.00
B / 1	5	8	2,026.00	2,030.00	2,028.00	2,028.00	4.00
B / 2	1	16	2,034.00	2,037.00	2,035.00	2,035.00	3.00
B / 2	50	16	2,039.00	2,042.00	2,040.00	2,040.00	3.00
B / 2	5	16	2,035.00	2,038.00	2,036.00	2,036.00	3.00
B / 2	1	8	2,034.00	2,038.00	2,036.00	2,036.00	4.00
B / 2	5	8	2,035.00	2,039.00	2,037.00	2,037.00	4.00
B / 3	1	16	2,025.00	2,029.00	2,027.00	2,027.00	4.00
B / 3	50	16	2,030.00	2,033.00	2,031.00	2,031.00	3.00
B / 3	5	16	2,026.00	2,029.00	2,027.00	2,027.00	3.00
B / 3	1	8	2,026.00	2,030.00	2,028.00	2,028.00	4.00
B / 3	5	8	2,026.00	2,030.00	2,028.00	2,028.00	4.00
B / 4	1	16	2,024.00	2,031.00	2,029.00	2,029.00	7.00
B / 4	50	16	2,040.00	2,043.00	2,041.00	2,041.00	3.00
B / 4	5	16	2,027.00	2,033.00	2,031.00	2,031.00	6.00
B / 4	1	8	2,025.00	2,034.00	2,032.00	2,032.00	9.00
B / 4	5	8	2,027.00	2,036.00	2,033.00	2,034.00	9.00

06. External MUX address order with equal phase slots

Purpose

Check whether a baseline pattern follows a physical external MUX address or changes when that address has a different predecessor.

What was done

1. Read all four ADC inputs on both arrays at the same six external MUX addresses 0-5, using each of five acquisition orders listed below.

2. After the six read slots, either keep the final address connected for one more slot or select external MUX address 7 without ADC conversions.

3. Plot the results sorted by physical external MUX address, irrespective of the order in which they were acquired.

Test parameters

Parameter	Setting
Orders	Ascending 0-1-2-3-4-5; descending 5-4-3-2-1-0; rotated 3-4-5-0-1-2; Gray 0-2-3-1-5-4; permuted 2-5-0-3-1-4.
Timing	d = 5 µs; six 100 µs read slots + one 100 µs boundary slot; period 710 µs.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

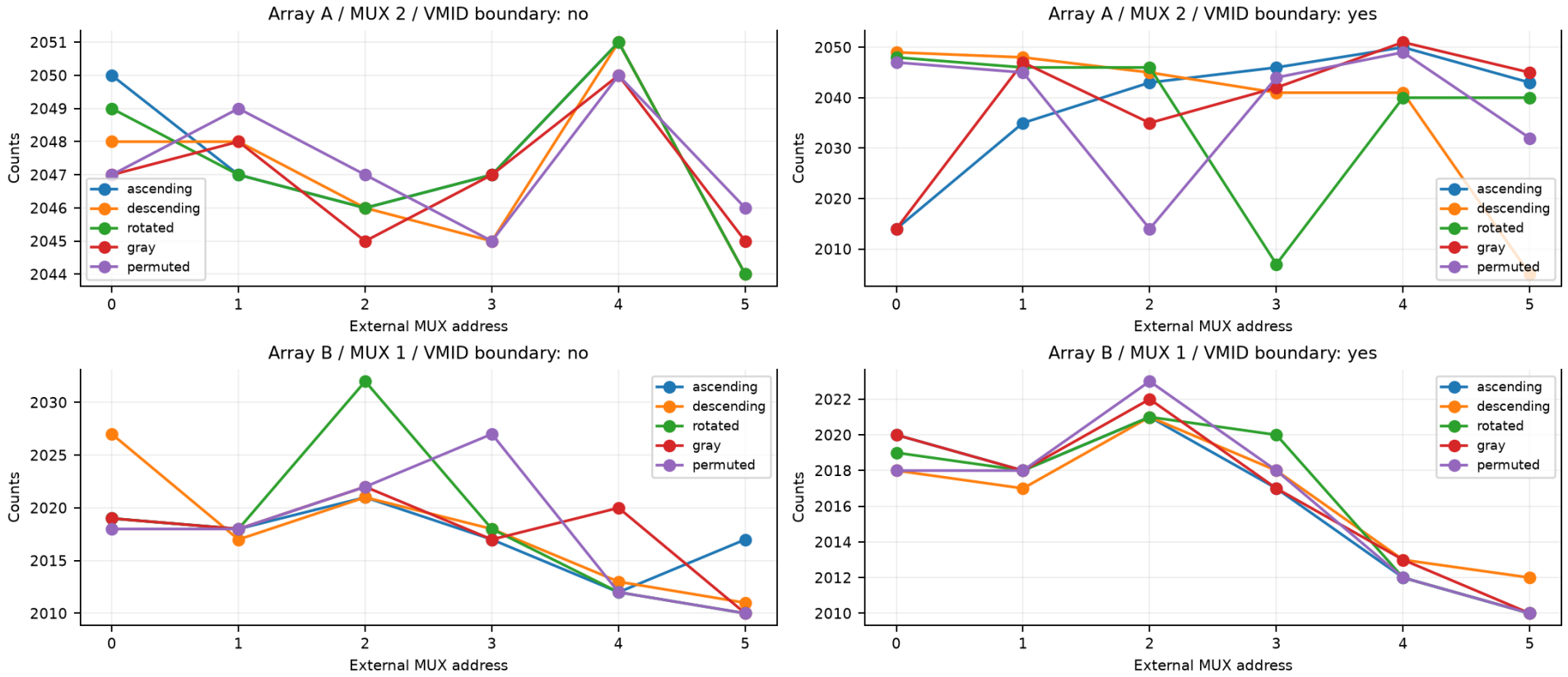
What was measured

Individual sensor baselines versus physical external MUX address. The figure uses Array A / MUX 2 and Array B / MUX 1 as examples; all paths are in channels.csv.

How to read the results

Colour = acquisition order, not a different sensor. X = physical external MUX address sorted 0-5. Left panels hold the last address at the boundary; right panels connect VMID there. Lines do not show chronological sample order.

06. External MUX address order - results



Conclusions from these results

Several baselines change with acquisition order. A peak at a physical address therefore cannot always be treated as a fixed sensor or resistor offset.

Gray order changes one address bit per transition, including wrap, but also changes the analog predecessor. These results do not separately identify GPIO feedthrough versus analog charge history.

07. Fast external MUX address order and placement of the cycle gap

Purpose

Test predecessor order and the location of the logging gap in a freely running sequence closer to the short-dwell scan.

What was done

1. Run full address sequences 0-1-2-3-4-5-6, 6-5-4-3-2-1-0 and 3-4-5-6-0-1-2, all with software VMID off.

2. Also run only addresses 0-5 in ascending or Gray order. Read IN1-IN4 at 0-5 and only IN4 at address 6.

3. Allow encoding/enqueue at each cycle end. Rotating the full sequence moves this gap from after address 6 to after address 2 while preserving cyclic address order.

Test parameters

Parameter	Setting
Timing	d = 5 µs; no fixed phase slots or cycle throttle. These are diagnostic, not production, sampling rates.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

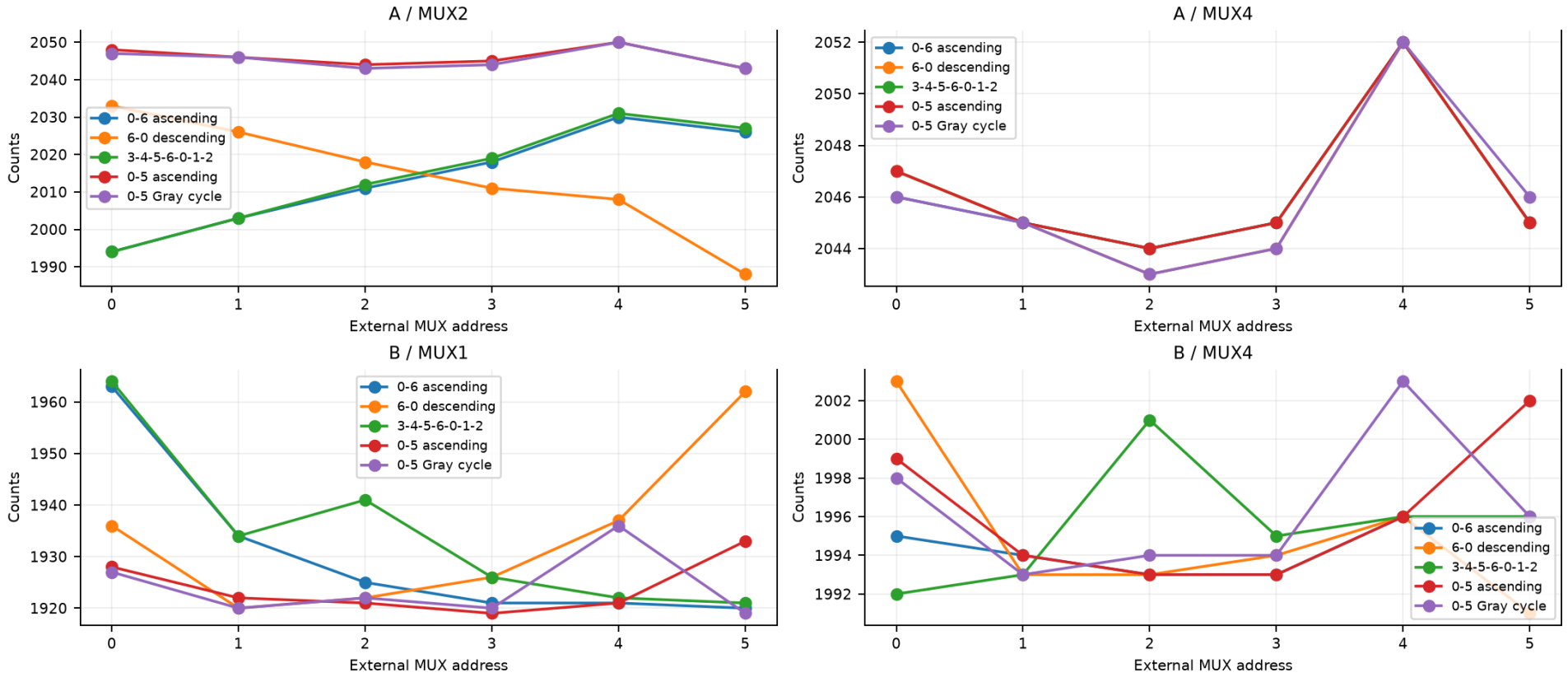
What was measured

Sensor baselines at the same external MUX addresses, with selected MUXes plotted. See the manifest for measured cycle periods.

How to read the results

Each line names the complete acquisition sequence. X is sorted physical external MUX address 0-5, not elapsed order. Address-6 readings are omitted from the plot so all curves compare the same sensors.

07. Fast order and cycle-gap placement - results



Conclusions from these results

Reversing order changes the address-dependent baseline profile. Rotating the sequence also changes some baselines, even though cyclic predecessors remain the same: where the longer cycle-boundary interval occurs matters.

The large Array B deficit persists. A different scan order is not by itself a demonstrated fix, and these graphs do not measure signal fidelity or ghosting.

08. ADC array order and ADC input order

Purpose

Check whether a path reads differently when it is converted earlier or later after the same external address selection.

What was done

1. Use external MUX addresses 0-5, with both arrays switching together. Add an address-7 boundary slot without conversions.

2. Compare ADC A then ADC B, each reading IN1-IN4; ADC B then ADC A, each IN1-IN4; and ADC A then ADC B, each reading IN4-IN1.

3. Keep the phase slots and cycle period equal across these conditions.

Test parameters

Parameter	Setting
Timing	Six 100 µs sensor slots + 100 µs VMID boundary; period 710 µs. Sensor wait d = 1, 5, 50 µs. Boundary wait 5 µs.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

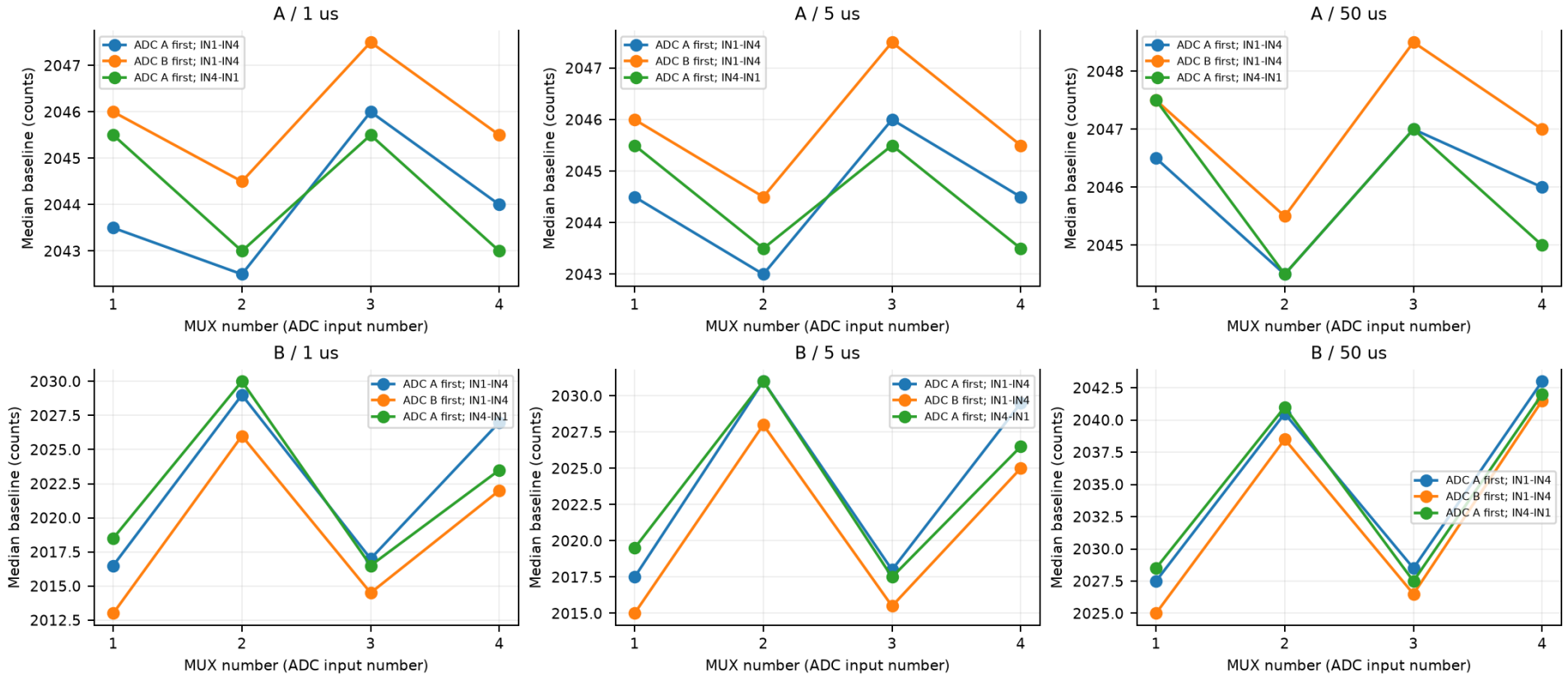
What was measured

For each MUX, the median baseline across external MUX addresses 0-5.

How to read the results

X = MUX number, also ADC input number; Y = baseline center. A-first/B-first says which ADC’s four-read burst occurs first. IN4-IN1 reverses ADC input order, not external MUX addresses.

08. ADC array and input order - results



Conclusions from these results

Swapping which ADC is read first changes baseline centers by up to 5 counts in these matched-slot tests. The ADC read second has additional elapsed time and preceding conversion activity.

Reversing ADC inputs changes both position in the burst and the preceding ADC input. These effects remain coupled; the result does not establish a universal direction or an independently measured optimal delay.

09. One active array versus both arrays

Purpose

Check whether activity in the other array affects readings beyond the simple delay caused by reading it first.

What was done

1. Use the same addresses and timing as Test 8. Record Array A alone, Array B alone and both arrays.

2. When only one array is active, leave the other array's external MUX EN disabled and do not read its ADC.

3. Compare the single-array result with the dual-array condition in which that same ADC is read first: A alone versus A-first, B alone versus B-first.

Test parameters

Parameter	Setting
Timing	Period 710 µs; six 100 µs sensor slots + 100 µs VMID boundary. d = 1, 5, 50 µs; ADC input order IN1-IN4.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

What was measured

MUX baseline centers and the difference "one active array minus both arrays with this ADC first."

How to read the results

The difference column is in counts. This matched-first comparison reduces the extra-wait difference. It does not make all bus, GPIO or reference-current activity identical.

09. One active array versus both arrays - results

Array / MUX	d (µs)	Both; ADC A first	Both; ADC B first	Only this array	Only - matched-first both
A / 1	1	2,043.50	2,046.00	2,044.00	0.50
A / 1	5	2,044.50	2,046.00	2,044.50	0.00
A / 1	50	2,046.50	2,047.50	2,046.50	0.00
A / 2	1	2,042.50	2,044.50	2,043.00	0.50
A / 2	5	2,043.00	2,044.50	2,043.50	0.50
A / 2	50	2,044.50	2,045.50	2,044.50	0.00
A / 3	1	2,046.00	2,047.50	2,046.00	0.00
A / 3	5	2,046.00	2,047.50	2,046.50	0.50
A / 3	50	2,047.00	2,048.50	2,047.50	0.50
A / 4	1	2,044.00	2,045.50	2,044.50	0.50
A / 4	5	2,044.50	2,045.50	2,044.50	0.00
A / 4	50	2,046.00	2,047.00	2,046.00	0.00
B / 1	1	2,016.50	2,013.00	2,013.00	0.00
B / 1	5	2,017.50	2,015.00	2,015.00	0.00
B / 1	50	2,027.50	2,025.00	2,025.50	0.50
B / 2	1	2,029.00	2,026.00	2,026.00	0.00
B / 2	5	2,031.00	2,028.00	2,028.00	0.00
B / 2	50	2,040.50	2,038.50	2,039.00	0.50
B / 3	1	2,017.00	2,014.50	2,014.50	0.00
B / 3	5	2,018.00	2,015.50	2,015.50	0.00
B / 3	50	2,028.50	2,026.50	2,026.50	0.00
B / 4	1	2,027.00	2,022.00	2,022.00	0.00
B / 4	5	2,029.50	2,025.00	2,025.00	0.00
B / 4	50	2,043.00	2,041.50	2,041.50	0.00

Conclusions from these results

The matched-first differences are at most 0.5 count in this schedule. This reduces the case for a large independent penalty simply from having the other array active. It does not remove the shared conversion-history effect demonstrated directly in Test 14. Single-array GPIO handling also differs slightly.

10. VMID connection versus extra conversions: 50 µs interval

Purpose

Determine whether an analog VMID connection changes baselines, and whether reading/discarding ADC values during that same connection adds an effect.

What was done

1. At external MUX address 0, wait d and read IN1-IN4 on ADC A then ADC B. Keep the sensor phase 100 µs long, then disable both external MUX banks.

2. Use the next 50 µs phase in one of three states: stay disabled; select address 7 without conversions; or select address 7, wait 5 µs and read/discard four conversions per ADC. Disable at phase end.

3. Repeat that sensor/reference pair for external MUX addresses 1-5. Thus the reference or disabled interval occurs after every address, not just at the cycle boundary.

Test parameters

Parameter	Setting
Timing	Sensor slot 100 µs; reference/disabled slot 50 µs; cycle period 920 µs; d = 1, 5, 50 µs.
EN and wait	Every sensor slot ends with EN disabled. Every reference slot also ends disabled. VMID wait is fixed at 5 µs; it does not follow d in this test.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

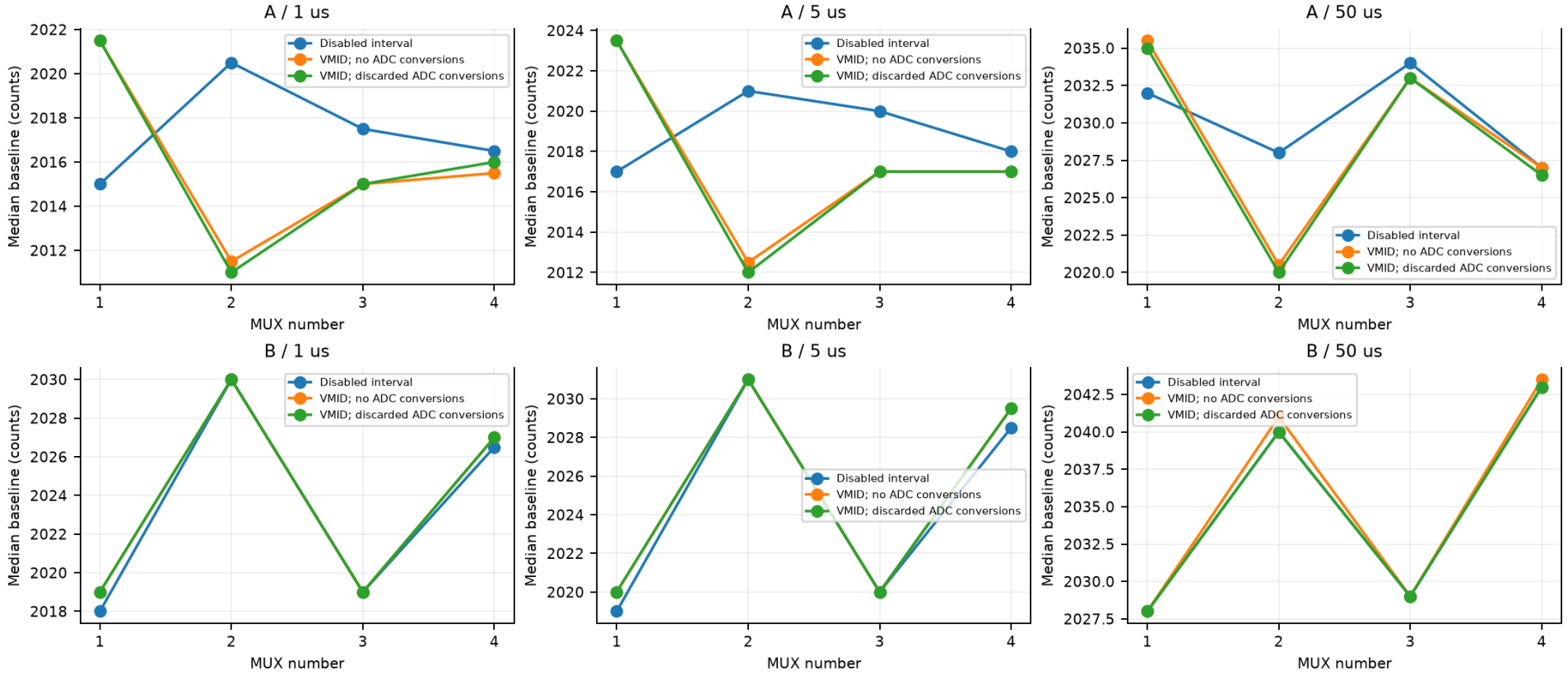
What was measured

Median sensor baseline across external MUX addresses 0-5, for each MUX. Only sensor reads contribute to the plotted baselines.

How to read the results

Blue “Disabled interval” = no selected external input between sensor slots. Orange “VMID; no ADC” = analog address-7 connection. Green “VMID + discarded ADC” = same reference connection with conversion activity. X = MUX number; columns = sensor wait d.

10. VMID connection versus extra conversions: 50 µs interval - results



Conclusions from these results

The two VMID variants nearly overlap: adding conversions during the otherwise equal connection changes median MUX centers by at most 1 count across this test's conditions.

The analog VMID connection itself can change baselines, particularly on Array A, and does not uniformly move every path closer to 2048. These controlled 50 µs intervals do not prove that production VMID conversions can be removed safely at a shorter dwell or faster cadence.

11. VMID connection versus extra conversions: 20 µs interval

Purpose

Determine whether an analog VMID connection changes baselines, and whether reading/discarding ADC values during that same connection adds an effect.

What was done

1. At external MUX address 0, wait d and read IN1-IN4 on ADC A then ADC B. Keep the sensor phase 25 µs long, then disable both external MUX banks.

2. Use the next 20 µs phase in one of three states: stay disabled; select address 7 without conversions; or select address 7, wait 5 µs and read/discard four conversions per ADC. Disable at phase end.

3. Repeat that sensor/reference pair for external MUX addresses 1-5. Thus the reference or disabled interval occurs after every address, not just at the cycle boundary.

Test parameters

Parameter	Setting
Timing	Sensor slot 25 µs; reference/disabled slot 20 µs; cycle period 300 µs; d = 1, 5 µs.
EN and wait	Every sensor slot ends with EN disabled. Every reference slot also ends disabled. VMID wait is fixed at 5 µs; it does not follow d in this test.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

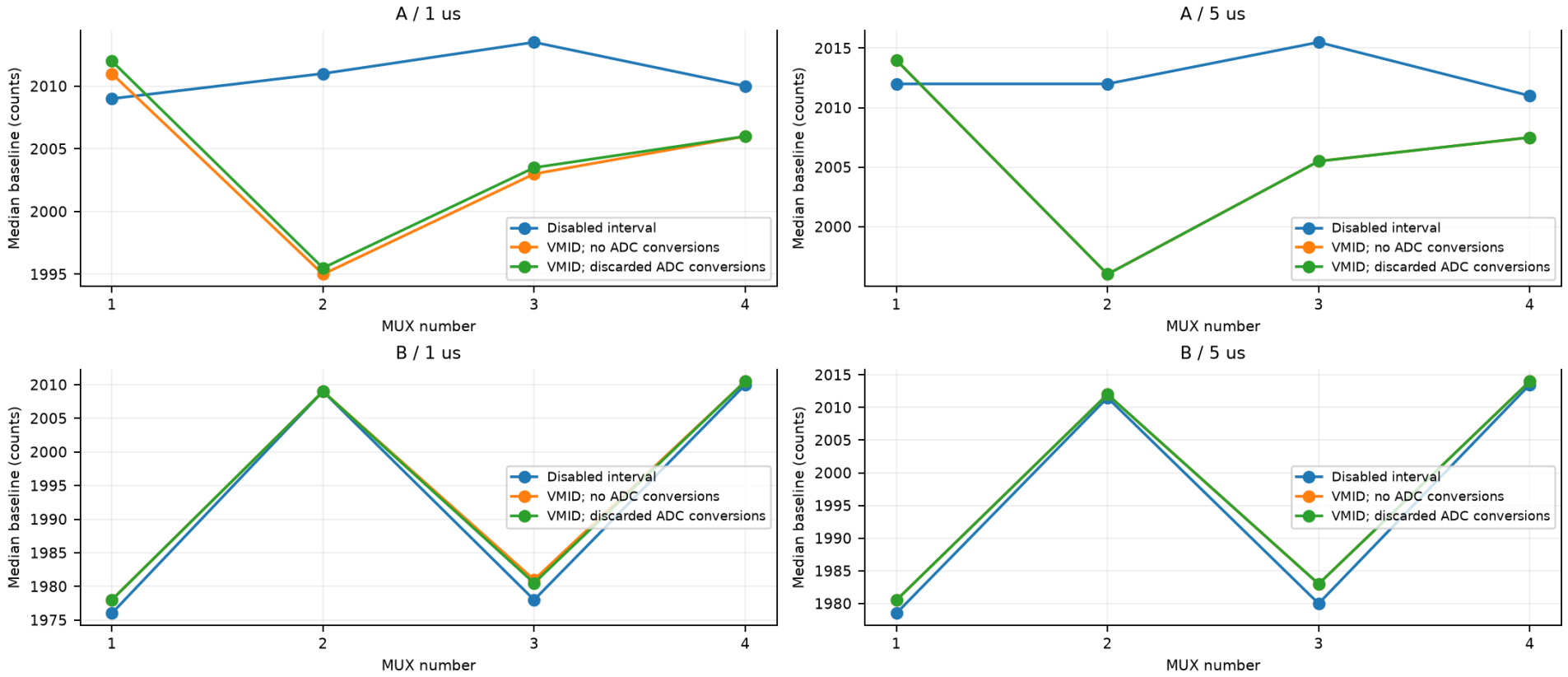
What was measured

Median sensor baseline across external MUX addresses 0-5, for each MUX. Only sensor reads contribute to the plotted baselines.

How to read the results

Blue “Disabled interval” = no selected external input between sensor slots. Orange “VMID; no ADC” = analog address-7 connection. Green “VMID + discarded ADC” = same reference connection with conversion activity. X = MUX number; columns = sensor wait d.

11. VMID connection versus extra conversions: 20 µs interval - results



Conclusions from these results

The two VMID variants nearly overlap: adding conversions during the otherwise equal connection changes median MUX centers by at most 1 count across this test's conditions.

The analog VMID connection itself can change baselines, particularly on Array A, and does not uniformly move every path closer to 2048. These controlled 20 µs intervals do not prove that production VMID conversions can be removed safely at a shorter dwell or faster cadence.

12. MUX isolation and reconnection at an unchanged address

Purpose

Check how a longer disconnected interval affects the next sensor reading, without requiring an external address-bit change at reconnection.

What was done

1. Use one array and one ADC input per recording; disable the other array. Connect external MUX address 2 for a 100.1 ms preconditioning phase without ADC reads.

2. Disable the active array’s external MUXes while keeping their cached address at 2. Wait the programmed interval, then finish a slot that is another 100 µs longer.

3. Re-enable at the same external MUX address 2, wait 1 µs and retain eight consecutive conversions of the same ADC input. Repeat the whole cycle.

Test parameters

Parameter	Setting
Programmed off-waits	1, 10, 100, 1000 ms. Actual disabled slots are approximately 1.1, 10.1, 100.1, 1000.1 ms, plus small control overhead.
Cycle / recordings	Period = programmed off-wait + 100.5 ms. Read slot 100 µs. Three recordings per path/interval; at least six cycles each; no initial warm-up exclusion.
SPI / initial condition	LPSPI, requested 16 MHz. Pre-disconnection sensor voltage was not recorded; complete charging was not established.

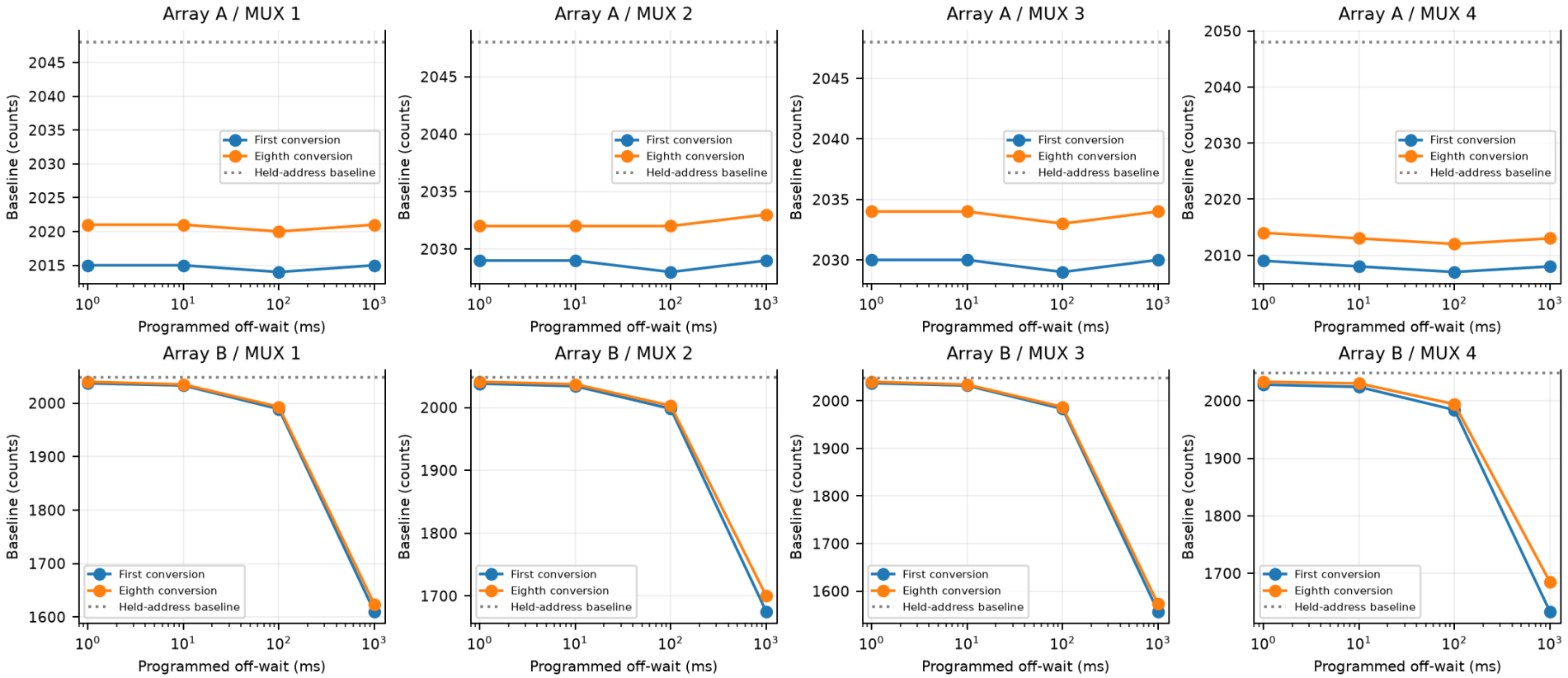
What was measured

First and eighth code on reconnection. The comparison is an early reconnect response, not a direct voltage measurement while the sensor is disconnected.

How to read the results

X = programmed off-wait in ms on a logarithmic axis; the actual disabled slot adds 0.1 ms. Blue = first conversion; orange = eighth. Grey dotted line = separate continuously held-address baseline.

12. Isolation and reconnection - results



Conclusions from these results

Between the shortest and longest programmed off-waits, first readings change by -1 to 0 counts on Array A, versus -480 to -363 counts on Array B (about -387 to -292 mV). This is a strong array-dependent retention/reconnection difference.

Address pins are unchanged at reconnection, so an address-bit transition is not required for the early-reading deficit. EN effects, sensor/MUX/PCB leakage, charge sharing and ADC history remain combined. These data do not identify a leakage resistance or sensor capacitance.

12. Isolation and reconnection - endpoint table

Column labels use the programmed wait; actual disabled slots add about 0.1 ms. Values are median codes across three recordings. The last column shows repetition variation, not temporal noise within a recording.

Array / MUX	First, 1 ms wait	First, 1000 ms wait	Change counts / mV	Eighth, 1000 ms wait	First-code range across reps
A / 1	2,015.00	2,015.00	0.00 / 0.00	2,021.00	3.00
A / 2	2,029.00	2,029.00	0.00 / 0.00	2,033.00	0.00
A / 3	2,030.00	2,030.00	0.00 / 0.00	2,034.00	0.00
A / 4	2,009.00	2,008.00	-1.00 / -0.81	2,013.00	2.00
B / 1	2,037.00	1,610.00	-427.00 / -344.02	1,624.00	8.50
B / 2	2,038.00	1,675.00	-363.00 / -292.46	1,700.00	2.50
B / 3	2,037.00	1,557.00	-480.00 / -386.72	1,573.00	6.00
B / 4	2,028.00	1,633.50	-394.50 / -317.83	1,685.00	41.50

Between cycles the input remains connected during the read-slot remainder, cycle padding and the next preconditioning phase. The experiment supplies a repeatable periodic history, not a guaranteed fully charged sensor before every off interval.

13. Known-voltage comparison of external MUX addresses 6 and 7

Purpose

Check readings while selecting two different external input pins that are physically connected to the same VMID net on MUXes 1-3.

What was done

1. Read only ADC IN1-IN3. In separate recordings, hold external MUX address 6, hold address 7, or alternate 6 then 7.

2. Use both arrays at delays 1, 5 and 50 µs. Add single-array alternating controls at 5 µs.

3. Do not read ADC IN4: its external MUX address 6 is a sensor, not the reference. MUX 4 still follows the shared address pins physically.

Test parameters

Parameter	Setting
Timing	100 µs phase slots; period 710 µs. Held cases read three inputs per ADC once per cycle; alternating cases read them twice.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

What was measured

Baseline at addresses 6 and 7 and raw SD while alternating. Nominal VMID is 2048 counts.

How to read the results

Table rows identify array, MUX number and settling delay. Both alternating columns refer to the same physical reference voltage reached through different external MUX input pins.

13. Known-voltage control - results

d (µs)	Array / MUX	Hold address 6	Hold address 7	Alternate: address 6	Alternate: address 7	Alternate SD: 6 / 7
1	A / 1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.017
1	A / 2	2,048.00	2,048.00	2,048.00	2,048.00	0.095 / 0.045
1	A / 3	2,049.00	2,049.00	2,049.00	2,049.00	0.383 / 0.481
1	B / 1	2,049.00	2,049.00	2,049.00	2,049.00	0.024 / 0.073
1	B / 2	2,049.00	2,049.00	2,049.00	2,049.00	0.034 / 0.000
1	B / 3	2,050.00	2,050.00	2,049.00	2,049.00	0.487 / 0.344
5	A / 1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.000
5	A / 2	2,048.00	2,048.00	2,048.00	2,048.00	0.135 / 0.067
5	A / 3	2,049.00	2,049.00	2,049.00	2,049.00	0.345 / 0.452
5	B / 1	2,049.00	2,049.00	2,049.00	2,049.00	0.000 / 0.053
5	B / 2	2,049.00	2,049.00	2,049.00	2,049.00	0.041 / 0.000
5	B / 3	2,050.00	2,050.00	2,049.00	2,049.00	0.497 / 0.376
50	A / 1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.000
50	A / 2	2,048.00	2,048.00	2,048.00	2,048.00	0.156 / 0.075
50	A / 3	2,049.00	2,049.00	2,049.00	2,049.00	0.312 / 0.446
50	B / 1	2,049.00	2,049.00	2,049.00	2,049.00	0.000 / 0.067
50	B / 2	2,049.00	2,049.00	2,049.00	2,049.00	0.029 / 0.000
50	B / 3	2,050.00	2,050.00	2,049.00	2,049.00	0.497 / 0.357

Conclusions from these results

These known-reference connections read about 2048-2050 counts. The large sensor-scan deficits do not appear under this low-impedance reference condition. One- or two-count differences remain dependent on timing and read position. Because MUX 4 still moves between its sensor and VMID, this is not a pure isolated GPIO-feedthrough measurement.

14. Repeated conversions while continuously connected to VMID

Purpose

Check whether an eight-reading change can occur without either external MUX address switching, ADC input switching or sensor charging.

What was done

1. Hold external MUX address 7 continuously on the active array or arrays. Keep the ADC input selection fixed at IN1.

2. Every 250 µs, use a 120 µs no-read reference phase, then read IN1 eight times in a 100 µs slot. The read phase includes a 1 µs wait.

3. Compare Array A only, Array B only, both arrays with ADC A's eight reads first, and both with ADC B's eight reads first. In single-array runs, disable the other array's MUXes.

4. Also record one conversion per active ADC per cycle; the graph's dotted reference uses the corresponding single-array one-read run.

Test parameters

Parameter	Setting
Timing	Period 250 µs; no-read phase includes a 100 µs wait in a 120 µs slot; read phase 100 µs. Requested 16 MHz.
SPI / arrays	LPSPI; requested 16 MHz; both arrays unless stated otherwise.
Recordings	Three recordings per condition; about 3.5 s each. Exclude the first 1 s; analyze the following 2.5 s.

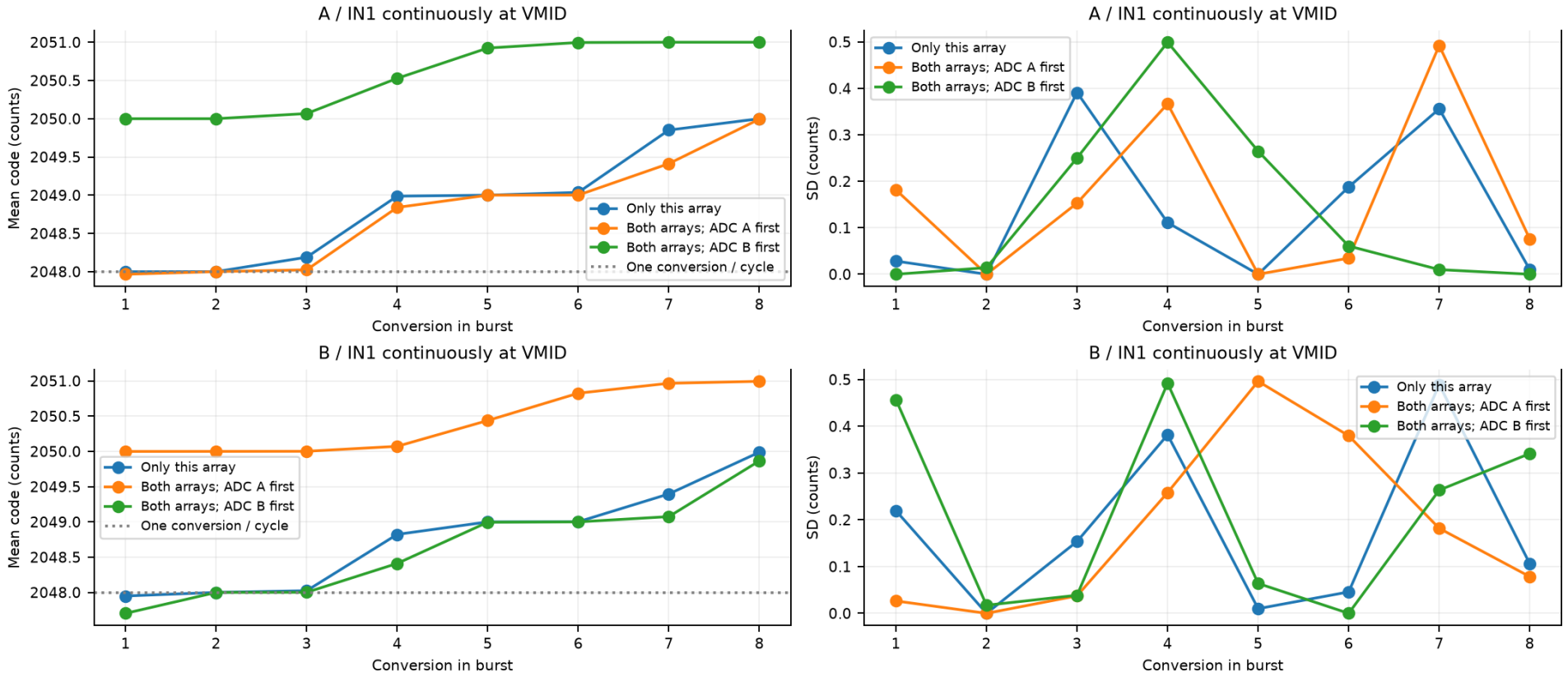
What was measured

Mean code and temporal SD separately for each conversion position. Means reveal sub-count changes in code occupancy; plotted values are medians of three per-record means/SDs.

How to read the results

X = conversion position within this ADC's burst. Blue = only this array; orange = both, ADC A first; green = both, ADC B first. Grey dotted baseline in the mean panels = one conversion/cycle with this array alone.

14. Continuously connected VMID - results



Conclusions from these results

The first-read ADC rises by about two counts across its eight conversions. The ADC read second starts about two counts higher; reversing array order transfers that behaviour to the other ADC.

A shared conversion-history effect contributes even without sensor or address transitions. Reference/supply changes, buffer drive, ADC behaviour and digital coupling remain candidates; this is not proof of a failed ADC or a requirement to discard a fixed number of samples.

14. Reference-only endpoints and what a code change could mean

Activity	Array	First mean	Eighth mean	Eighth - first counts
A only	A	2,047.9992	2,049.9999	2.0007
B only	B	2,047.9493	2,049.9886	2.0393
A then B	A	2,047.9659	2,049.9942	2.0283
A then B	B	2,049.9993	2,050.9938	0.9945
B then A	B	2,047.7033	2,049.8651	2.1618
B then A	A	2,050.0000	2,051.0000	1.0000

Interpretation of the reference ratio

Near midscale, the ADC code is approximately 4096 × input voltage / reference voltage. A code rise can therefore come from the buffered VMID input rising, the ADC reference falling, or both. The ADC uses VA as its analog supply and reference.

For scale only: +3 counts could mean the input rose about 2.42 mV with a fixed 3.3 V reference, or the reference fell about 4.83 mV with the input fixed at 1.65 V. These are equivalent-voltage calculations, not measured rail changes. Shared internal reference generation does not guarantee that fast output-load disturbances cancel.

Simultaneous probing of ADC reference/supply, VMID and buffer output is needed to distinguish those candidates. The same code shift can have several electrical explanations.

15. Long recordings: time behaviour and sustained disturbances

Purpose

Look for drifting baselines, bursts and steps that would not be represented by a single SD number.

What was done

1. Record addresses 0-5 with no explicit VMID: six 100 µs read slots. The last address 5 remains connected for the rest of the 1 ms cycle.

2. Record addresses 0-5 with VMID after every address: alternate 100 µs sensor slots and 50 µs address-7 slots; read/discard four conversions per ADC during each VMID slot. VMID remains connected during the cycle remainder.

3. Record with external MUX address 2 held continuously: one four-input read burst per ADC every 1 ms. Keep the external connection unchanged between reads.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.

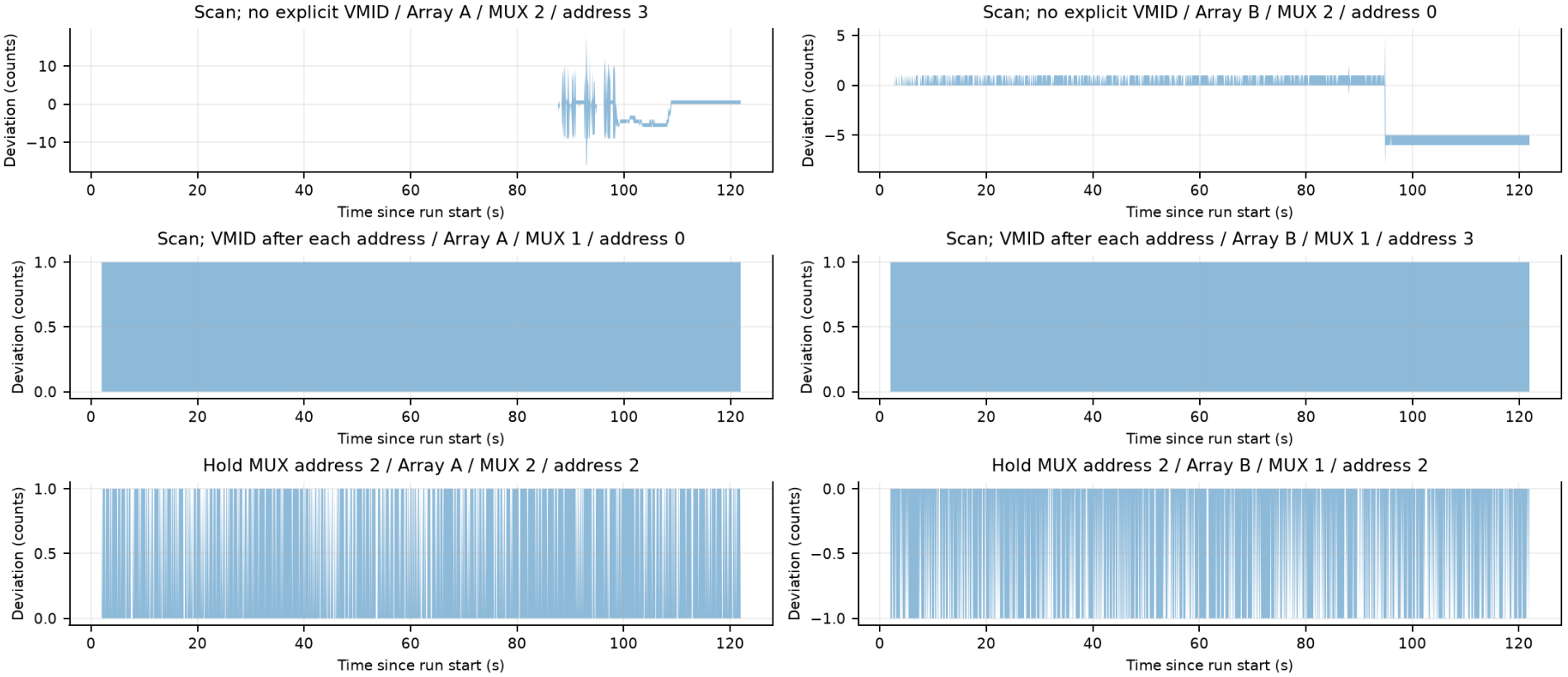
What was measured

For each array and condition, select the sensor route with the largest raw SD. Plot all analyzed data as minimum/maximum envelopes over successive 100 ms bins.

How to read the results

Rows = the three conditions above; columns = arrays. Each panel identifies its selected MUX number and external MUX address. Y = code minus that route’s full-record median. Blue shading covers each bin’s minimum to maximum; it is not a confidence interval or averaged trace.

15. Long-record time traces - results



Conclusions from these results

The no-explicit-VMID record contains a late burst and a baseline step. It is not adequately described as stationary white noise.

The later VMID and held records have less sustained variation, but these are separate times and one recording per condition. The difference does not establish the cause or prove that VMID suppresses disturbances. Selecting by SD can miss a rare large spike; Test 16 therefore selects independently by range.

16. Long recordings: isolated spikes and maximum excursions

Purpose

Find large brief excursions even when they contribute too little to make their route the highest-SD example.

What was done

1. Use all three existing long recordings from Test 15; do not remove spikes.

2. For each array and condition, choose the sensor route with the largest full-record peak-to-peak range, independently of the SD selection.

3. Find its sample furthest from the median and plot every recorded point in a 25 ms window around that event.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.

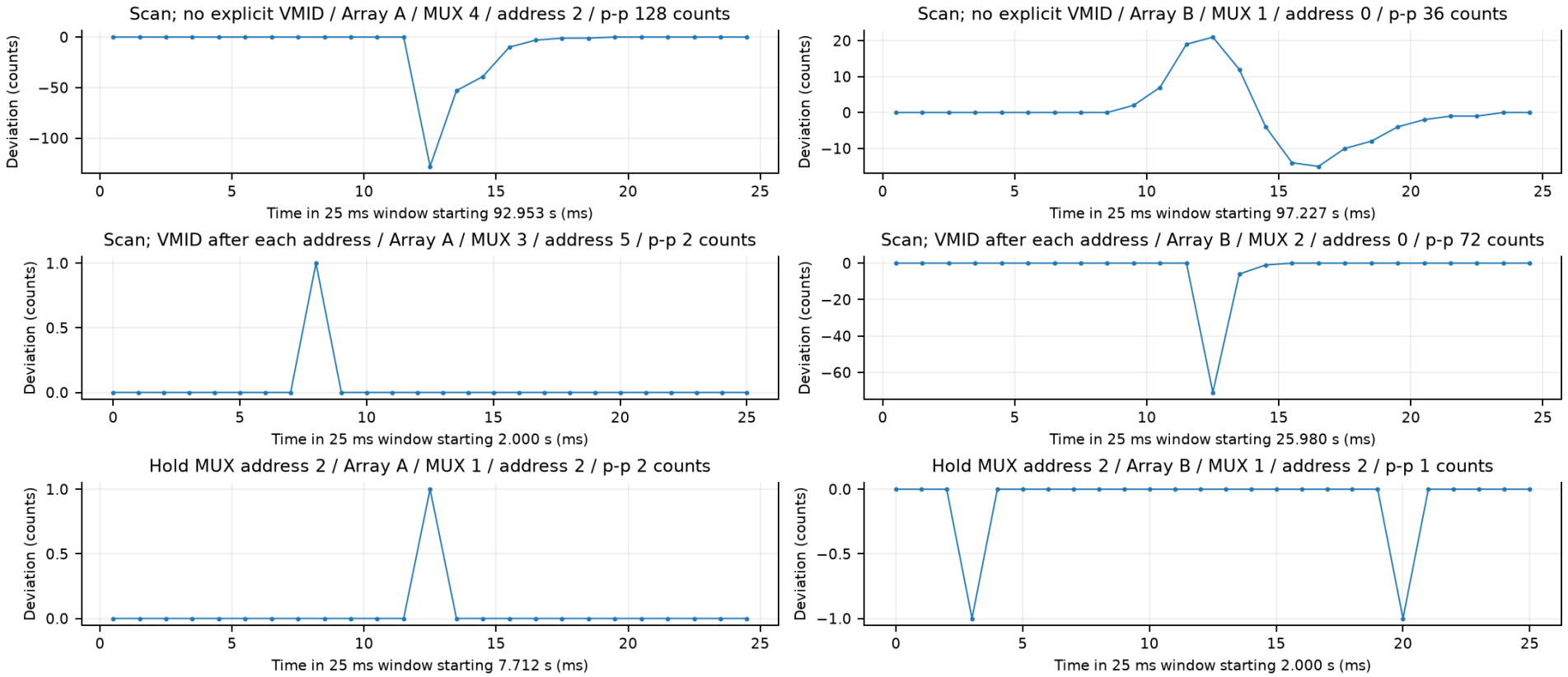
What was measured

Full-record peak-to-peak and raw SD for the selected route, in counts and nominal mV, plus the event’s sampled time shape.

How to read the results

Dots are individual samples, about 1 ms apart; connecting lines are visual aids. Y = deviation from median. X = milliseconds within the 25 ms window; the starting time since acquisition began appears below each panel. The labelled peak-to-peak is the full-record range, not necessarily the span visible in that one window.

16. Largest excursions - results



Conclusions from these results

The largest route ranges are 128 counts (103.1 mV) in the no-explicit-VMID scan, 72 counts (58.0 mV) with VMID, and 2 counts held. The VMID condition still contains a brief large spike despite low sustained SD.

Valid framing confirms these are acquired ADC codes rather than undecoded transport bytes. It does not distinguish electrical interference from a real PZT response to vibration.

16. Largest excursions - numerical comparison

Condition	Array / MUX / external address	Raw range counts / mV	Raw SD counts / mV
Scan; no explicit VMID	A / MUX 4 / address 2	128 / 103.125	0.4387 / 0.3535
Scan; no explicit VMID	B / MUX 1 / address 0	36 / 29.004	0.1538 / 0.1239
Scan; VMID after each address	A / MUX 3 / address 5	2 / 1.611	0.0835 / 0.0673
Scan; VMID after each address	B / MUX 2 / address 0	72 / 58.008	0.2059 / 0.1659
Hold MUX address 2	A / MUX 1 / address 2	2 / 1.611	0.0104 / 0.0084
Hold MUX address 2	B / MUX 1 / address 2	1 / 0.806	0.1144 / 0.0922

A low SD does not guarantee no rare excursions. Conversely, one-count toggling is expected when an analog level lies near an ADC code boundary. The sample rate limits the event shape visible here; sub-millisecond behaviour was not captured.

17. Long recordings: power spectral density

Purpose

Show whether fluctuations are broadly distributed with frequency or concentrated at low frequencies or particular frequencies.

What was done

1. Use the same highest-SD routes as Test 15, with every steady-state sample retained for the raw curve.

2. Separately identify the longest continuous quiet interval common to all routes in that recording, using the screening method in Test 18.

3. Compute Welch spectra using up to 2048 samples per window, subtracting the window mean. Use only uninterrupted received sequences; do not interpolate across missing cycles.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.
Spectrum	About 1 ksample/s per route; frequency axis extends to about 500 Hz. Constant-detrended Welch windows, up to 2048 points.

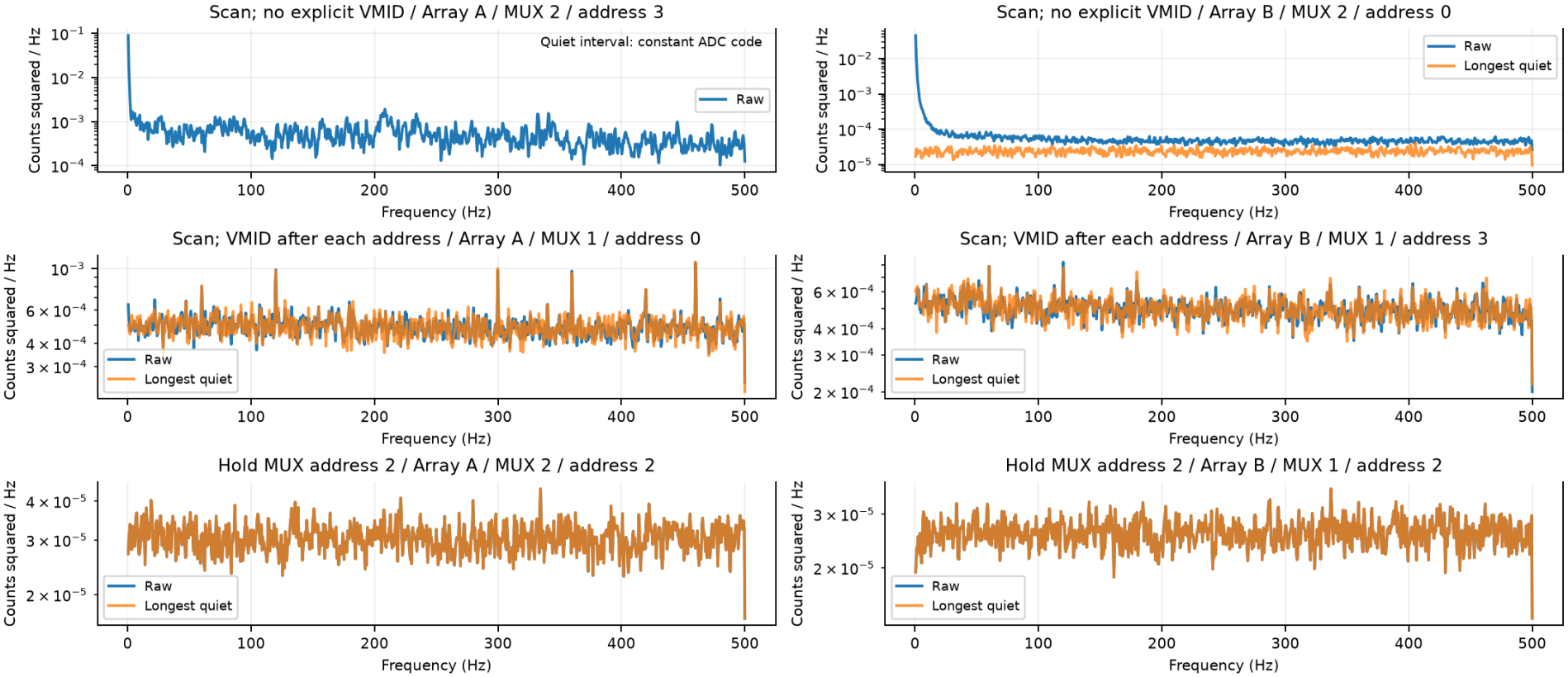
What was measured

Power spectral density in counts squared per Hz, for the full raw record and the selected quiet interval.

How to read the results

Blue = raw; orange = longest quiet interval. X = frequency in Hz; Y is logarithmic PSD. A flat spectrum can be consistent with white noise, but a PSD alone does not prove whiteness or identify a source. A constant-code quiet interval has zero recorded PSD and is annotated instead of drawn at a fictitious floor.

17. Power spectral density - results



Conclusions from these results

The no-explicit-VMID burst/step contributes strong low-frequency power, especially on Array B. Removing sustained disturbed intervals changes that spectrum.

Some remaining spectra are broad and low, but isolated events, quantization and limited duration prevent declaring all inputs purely white-noise sources. The displayed routes are selected examples; full raw/quiet route statistics are exported.

18. Long recordings: raw versus quiet noise statistics

Purpose

Report ordinary quiet behaviour separately from full-record behaviour while keeping spikes visible in the raw statistics.

What was done

1. Compute raw SD and peak-to-peak from every steady-state sample in each long recording.

2. Screen coherent separated-code states and one-second periods of unusually high SD, using the detailed rules on the following page.

3. Take the longest remaining uninterrupted received interval shared by all routes. Compute quiet statistics on exactly that interval; do not join different baseline states or choose the lowest SD interval.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.

What was measured

Raw and quiet SD/range. The results table uses the highest-raw-SD route in each array/condition; all routes and exact intervals are in the companion CSV/JSON files.

How to read the results

Quiet means a defined screened time interval, not a guarantee of Gaussian or white noise. Isolated spikes inside that interval remain. A zero recorded SD means constant ADC codes, not zero physical analog noise.

18. Quiet-interval selection - exact rules

Stage	Rule
Separated-state screen	Anchor each route to a dominant one-code state (60% occupancy) or adjacent two-code state (80% occupancy), allowing a one-count margin. A separated excursion requires at least 1% tail mass and a histogram peak with at least 0.5% of samples and three times the edge density.
Common rejection	At least two routes must depart in the same cycle. Reject that cycle and one neighbouring cycle on each side, identically for every route.
Sustained-variance screen	For each route, compute SD in one-second bins. Flag a bin if its SD exceeds max(0.75 count, four times that route's median bin SD). Reject bins flagged by any route and pad by one bin on each side.
Continuous interval	From what remains after both screens, choose the longest uninterrupted received interval. Split at missing sequence numbers. Apply its same start/end to every route.
Audit files	long_quiet_metadata.json records thresholds and selected interval. long_quiet_channels.csv records every route's raw/quiet statistics. Raw records and raw statistics retain all events.

The method screens sustained or coherent disturbed states; it is not a complete detector for every isolated spike. Do not interpret quiet statistics as the worst-case excursions. Test 16 supplies the independent range check.

18. Raw and quiet noise statistics - results

Condition	Array / MUX / address	Raw SD counts / mV	Raw range counts / mV	Quiet SD counts / mV	Quiet range counts	Quiet interval s
Scan; no explicit VMID	A / 2 / 3	1.4227 / 1.1462	34 / 27.393	0.0000 / 0.0000	0	2.000 to 44.999
Scan; no explicit VMID	B / 2 / 0	2.1273 / 1.7139	13 / 10.474	0.1084 / 0.0873	1	2.000 to 44.999
Scan; VMID after each address	A / 1 / 0	0.4993 / 0.4023	1 / 0.806	0.4996 / 0.4025	1	27.000 to 121.999
Scan; VMID after each address	B / 1 / 3	0.4999 / 0.4028	1 / 0.806	0.5000 / 0.4028	1	27.000 to 121.999
Hold MUX address 2	A / 2 / 2	0.1242 / 0.1001	1 / 0.806	0.1242 / 0.1001	1	2.000 to 122.000
Hold MUX address 2	B / 1 / 2	0.1144 / 0.0922	1 / 0.806	0.1144 / 0.0922	1	2.000 to 122.000

Conclusions from these results

Quiet intervals can have much smaller SD than the full disturbed record. The chosen no-VMID Array-A route is constant-code in its quiet interval. The VMID examples near 0.5 count SD mostly toggle between adjacent codes; this is not by itself evidence of a 0.5-count continuous analog-noise floor.

These are SD-selected routes. For example, the 72-count VMID spike occurs on a different Array-B route and is therefore shown in Test 16, not hidden by this table’s small ranges.

19. Correlation between measured paths

Purpose

Look for fluctuations that occur together on different paths, while avoiding correlation caused only by a slow shared drift.

What was done

1. Use the long recordings from Test 15. Compare pairs acquired at the same external MUX address phase, requiring route SD above 0.05 count.

2. Compute ordinary correlation over the full steady-state record.

3. Subtract each route's own mean separately in each one-second bin and compute correlation again. Show the 14 pairs with largest absolute within-second correlation.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.

What was measured

Dimensionless Pearson correlation: +1 means rising/falling together; -1 means opposite directions; 0 means little linear association.

How to read the results

A route label such as A / MUX 2 / address 5 identifies the array, external MUX chip and external MUX address. Within-second means removing each one-second mean; it is not simultaneous sampling or a proof of coupling.

19. Shared correlations - results

Condition	Path 1	Path 2	Raw correlation	Within-second correlation
Scan; no explicit VMID	B / MUX 4 / address 5	B / MUX 1 / address 5	-0.515	-0.511
Scan; no explicit VMID	A / MUX 3 / address 2	A / MUX 2 / address 2	0.477	0.476
Scan; no explicit VMID	A / MUX 4 / address 2	A / MUX 2 / address 2	0.329	0.331
Scan; no explicit VMID	B / MUX 4 / address 1	B / MUX 2 / address 1	0.244	0.246
Scan; no explicit VMID	A / MUX 4 / address 2	A / MUX 3 / address 2	0.243	0.245
Scan; VMID after each address	A / MUX 3 / address 0	A / MUX 1 / address 0	0.233	0.223
Scan; VMID after each address	A / MUX 4 / address 0	A / MUX 1 / address 0	0.230	0.219
Scan; VMID after each address	A / MUX 4 / address 0	A / MUX 3 / address 0	0.218	0.205
Scan; VMID after each address	A / MUX 2 / address 0	A / MUX 1 / address 0	0.211	0.202
Scan; no explicit VMID	A / MUX 3 / address 5	A / MUX 1 / address 5	-0.375	-0.168
Scan; no explicit VMID	B / MUX 3 / address 2	B / MUX 1 / address 2	-0.145	-0.145
Scan; VMID after each address	A / MUX 3 / address 2	A / MUX 1 / address 2	0.109	0.109
Scan; no explicit VMID	A / MUX 3 / address 5	A / MUX 2 / address 5	0.081	0.089
Scan; VMID after each address	B / MUX 4 / address 1	B / MUX 3 / address 1	0.089	0.087

Conclusions from these results

Some pairs share measurable variation, including opposite-direction changes. Correlation narrows what to inspect, but does not distinguish a common reference disturbance, sensor/table motion or electrical coupling. Many pairs were screened, and ADC inputs were read sequentially; the largest values are descriptive, not a significance test.

20. Timing stability and association with code changes

Purpose

Check whether the long-record code events coincide with variable software delay or variable ADC-burst duration.

What was done

1. Use the same highest-SD routes as Test 15. Retrieve CPU-tick markers after MUX switching guards, immediately before the array's ADC burst and immediately after that burst.

2. Calculate wait-to-burst delay and full burst duration for each cycle, using integer tick differences before converting units.

3. Measure their total ranges and correlate each with the selected ADC code.

Test parameters

Parameter	Setting
Timing / SPI	LPSPI, requested 16 MHz; sensor wait 5 µs; explicit VMID wait 5 µs; cycle period 1000 µs; both arrays.
Recordings	One separate recording per condition: 2 s initial exclusion, then 120 s analyzed. Conditions occur at different times; no vibration monitoring.

What was measured

Median delay in µs, full-record timing range in ns, and code/timing correlation. The burst duration covers the array's whole read batch, not just the selected input.

How to read the results

A dash for correlation means the timing variable was constant, so correlation is undefined. These are firmware markers, not oscilloscope pin edges or exact sample-aperture times.

20. Timing stability - results

Condition / array	MUX / address	Median delay µs	Delay range ns	Burst-duration range ns	Code/delay correlation	Code/duration correlation
Scan; no explicit VMID / A	2 / 3	5.0767	5.00	5.00	0.0009	-0.0051
Scan; no explicit VMID / B	2 / 0	10.3433	10.00	1.67	-0.0024	-0.0015
Scan; VMID after each address / A	1 / 0	5.0767	5.00	13.33	0.0032	0.0044
Scan; VMID after each address / B	1 / 3	10.3417	11.67	10.00	0.0051	-0.0012
Hold MUX address 2 / A	2 / 2	5.0750	3.33	6.67	-0.0010	0.0010
Hold MUX address 2 / B	1 / 2	10.3433	6.67	6.67	-0.0003	-0.0020

Conclusions from these results

Logged timing is stable to a few or tens of nanoseconds, and its correlations with the selected codes are near zero. This does not support variable software burst timing as the main explanation for the large recorded events. Stable software timing does not rule out electrical USB, reference/supply or digital coupling.

21. One full cycle after a 20-second timed VMID hold

Purpose

Test how the first complete sensor scan after prolonged VMID parking differs from the immediately preceding continuous-scan baseline.

What was done

1. Scan all 50 sensors continuously with software VMID off. Read external MUX addresses 0-5 with ADC IN1-IN4, then address 6 with ADC IN4 only. Keep the final received full cycle as the before snapshot.

2. Stop and select external MUX address 7 on every MUX. Prepare the next ADC selector before starting the timed hold. Keep the analog VMID connection for 20 seconds without ADC conversions.

3. Switch to address 0 and acquire exactly one full address-0-to-6 cycle, with no explicit VMID visit between addresses. Stop and park again.

Test parameters

Parameter	Setting
Sensor wait / order	5 µs after each external address selection. ADC A IN1-IN4 first, then ADC B IN1-IN4; address 6 reads only IN4.
Hold timing	Timed no-conversion interval: 20.000 s. Total time from last received pre-cycle end to post-cycle start: 21.675-21.799 s. Host setup/selector preparation precede the timed interval.
SPI / arrays	LPSPI; requested 16 MHz; both arrays. External MUX address changes coordinated between arrays.
Before recording	3.5 s continuous full scan: exclude its first 1 s, analyze the last 2.5 s; also retain its last received cycle.
After recording	One cycle only after the timed hold; retain the first response, with no discarded sensor readings. Three repetitions per condition.

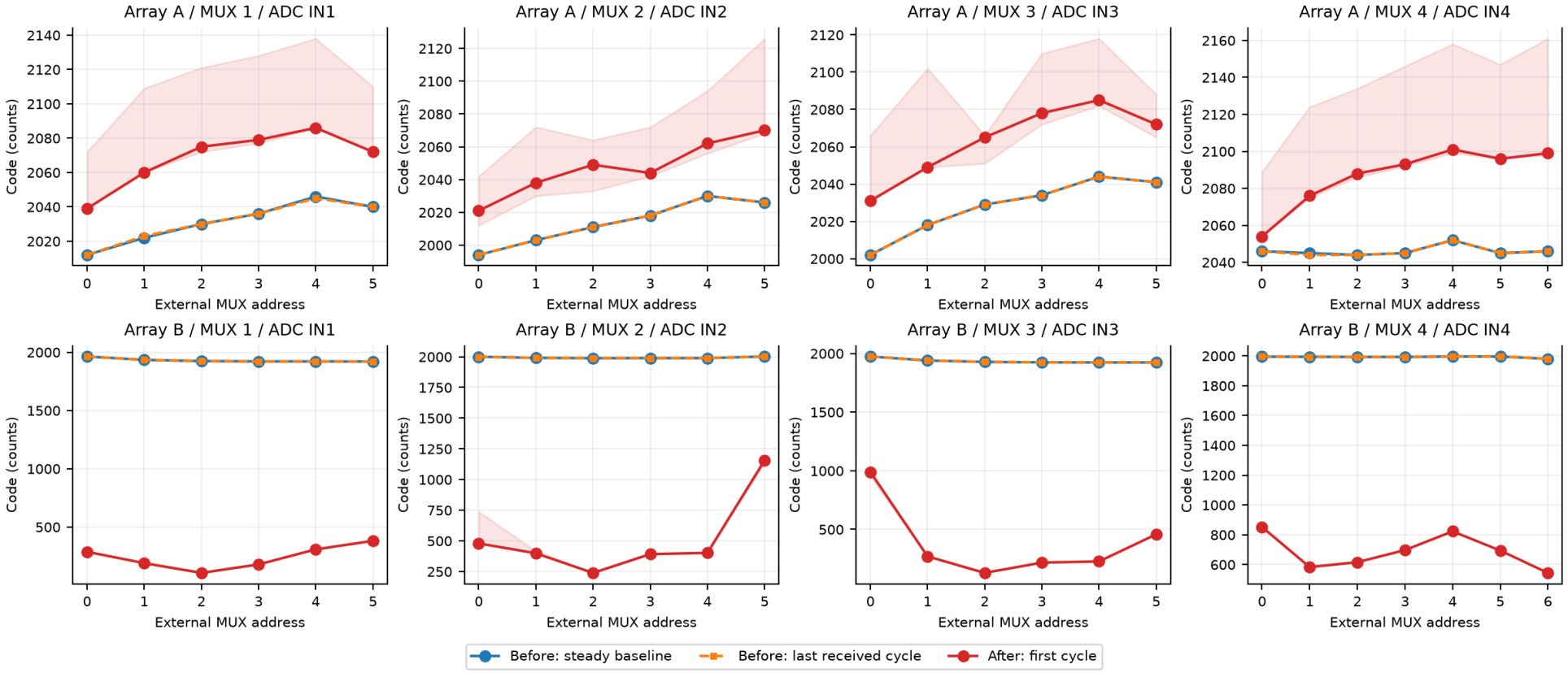
What was measured

For each physical sensor: the before-window baseline, its last received pre-cycle code, and its first post-hold-cycle code. The after value is a transient first reading, not a new steady baseline.

How to read the results

Each panel is one array and MUX number / ADC input. X = external MUX address. Blue = before-window baseline; orange dashed = last pre-cycle; red = first post-cycle. Curves are medians of three repetitions; red shading spans the minimum to maximum post code across those three repetitions, not noise SD.

21. First full cycle after the long hold - results



Conclusions from these results

Across external MUX addresses 0-5, MUX median changes are +33.5 to +46.0 counts on Array A, versus -1709.0 to -1299.0 on Array B. The result is not a uniform lowering on both arrays. The per-address graph shows which individual inputs depart from these summaries.

Parking on VMID also disconnects every sensor from its buffer/bias path for a long time. The effect therefore combines sensor off-time, retention/leakage and reconnection history with any reference-related effect. It does not isolate the 150 pF reference capacitor or establish charge accumulation between sensor addresses.

21. Full-cycle changes - numerical comparison

Only addresses 0-5 enter each MUX summary so all MUXes compare six sensors. MUX 4’s seventh sensor at address 6 is retained in the graph and route CSV. The change column is the median of matched sensor-route changes, which need not equal the difference of the two center columns.

Array / MUX	Before center counts / V	First-post center counts / V	Median route change counts / mV	Route-change range counts
A / 1	2,033.00 / 1.6379	2,073.50 / 1.6705	39.00 / 31.42	27.00 to 45.00
A / 2	2,014.50 / 1.6230	2,046.50 / 1.6488	33.50 / 26.99	26.00 to 44.00
A / 3	2,031.50 / 1.6367	2,068.50 / 1.6665	33.50 / 26.99	29.00 to 44.00
A / 4	2,045.00 / 1.6476	2,090.50 / 1.6842	46.00 / 37.06	8.00 to 51.00
B / 1	1,922.50 / 1.5489	238.50 / 0.1922	-1,709.00 / -1,376.88	-1,818.00 to -1,539.00
B / 2	1,990.00 / 1.6033	400.50 / 0.3227	-1,589.50 / -1,280.60	-1,749.00 to -847.00
B / 3	1,926.00 / 1.5517	248.50 / 0.2002	-1,683.00 / -1,355.93	-1,799.00 to -984.00
B / 4	1,994.50 / 1.6069	695.50 / 0.5603	-1,299.00 / -1,046.56	-1,411.00 to -1,143.00

A full post-hold cycle reads different sensors in succession. Its points are not the recharge trajectory of one sensor. Repeated-measurement variability is shown by the red band; it is not valid to calculate a noise SD from the 50 different sensor codes.

22. Ten conversions of one fixed ADC input after the long hold

Purpose

Measure the first ten responses on one sensor path after VMID parking, and check the influence of which ADC is read first.

What was done

1. Before each hold, record the same short continuous full scan as Test 21. Select external MUX address 2 as the target for the post-hold measurement.

2. Choose one MUX number m. Set the ADC selector to INm before the timed hold, with the external MUXes already parked at address 7. Then wait 20 seconds with no conversion clocks.

3. Switch both external MUX groups to address 2, wait 5 µs, and read INm ten consecutive times on ADC A, followed by ten on ADC B. Do not change external address or ADC input within either burst.

4. Repeat for MUX numbers 1, 2, 3 and 4. Repeat all four with ADC B's ten reads first, then ADC A's. Thus one address-2 sensor per MUX is tested on both arrays.

Test parameters

Parameter	Setting
Timing / selection	One post-hold external MUX address: 2. Sensor wait 5 µs. Ten retained readings per ADC, 20 codes total per post recording.
Read-order control	Before-scan ADC order matches its post-burst order. Changing order also changes prior full-scan read history slightly. The ADC read second has extra elapsed time and preceding conversions.
SPI / arrays	LPSPI; requested 16 MHz; both arrays. External MUX address changes coordinated between arrays.
Before recording	3.5 s continuous full scan: exclude its first 1 s, analyze the last 2.5 s; also retain its last received cycle.
After recording	One cycle only after the timed hold; retain the first response, with no discarded sensor readings. Three repetitions per condition.

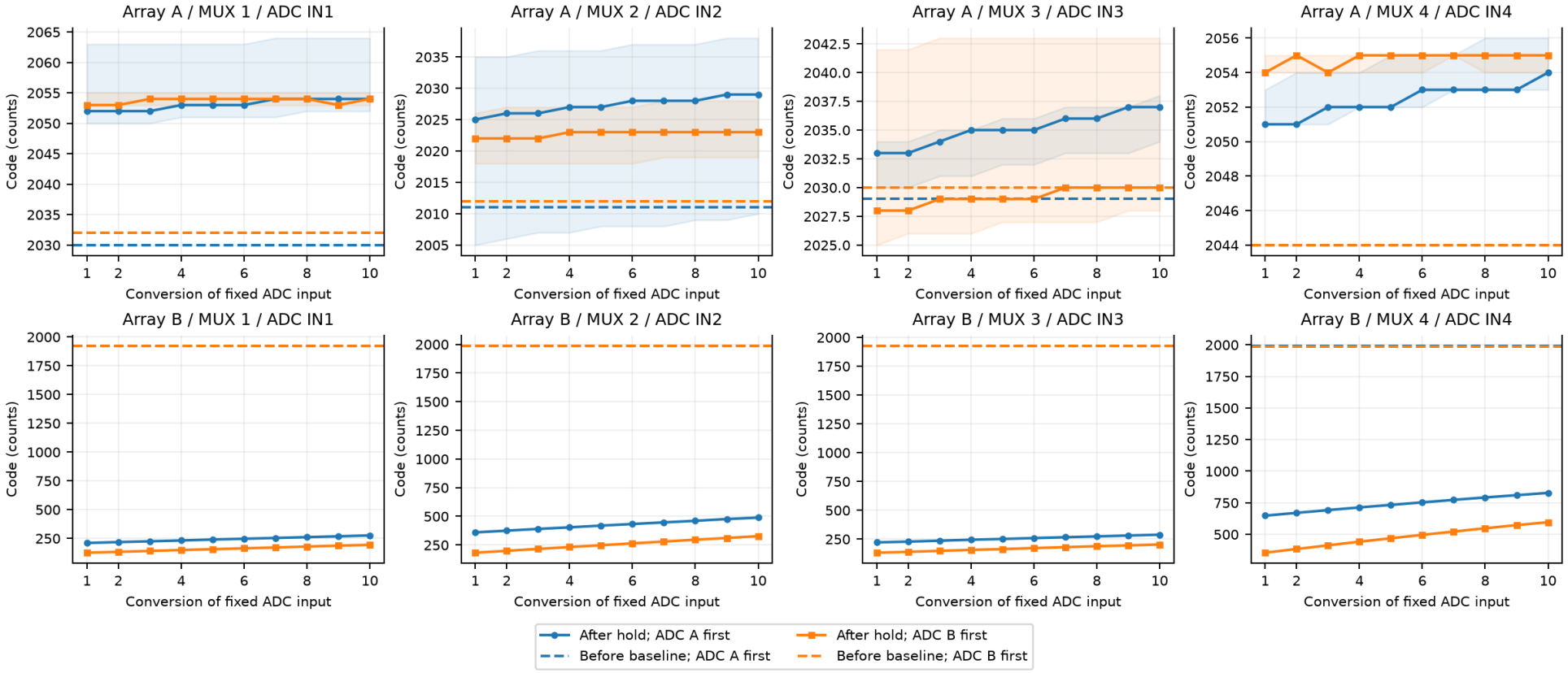
What was measured

The code at each conversion position 1-10, the before-window baseline for that same sensor, and the first-to-tenth change. These ten readings describe recovery, not stationary noise.

How to read the results

X = consecutive conversion position of the same ADC input. Blue circles = ADC A first; orange squares = ADC B first. Matching dashed lines are before-window baselines. Solid curves are median codes across three repeats; shading is the min-to-max repetition range, not SD or error bars.

22. Ten-conversion responses - results



Conclusions from these results

The median tenth-minus-first change spans +1.0 to +4.0 counts on Array A and +66.0 to +241.0 on Array B. The before-baseline dashed lines show whether ten readings return the sensor path to its prior scanning level.

Order differences combine additional settling time before the second ADC with shared conversion activity. They do not identify the reference as the only cause. The earlier continuously connected VMID control (Test 14) demonstrates a smaller conversion-history effect even without a sensor transition.

22. Array A - before baseline and burst endpoints

All targets are external MUX address 2. “First ADC” identifies the order of the two ten-reading bursts, not MUX number. The before baseline is from that same repetition’s continuous scan; reported values aggregate three repetitions.

MUX / ADC input	First ADC	Before baseline counts / V	First code counts / V	Tenth code counts / V	Tenth - first counts / mV
MUX 1 / IN1	A	2,030.00 / 1.6355	2,052.00 / 1.6532	2,054.00 / 1.6548	2.00 / 1.61
MUX 1 / IN1	B	2,032.00 / 1.6371	2,053.00 / 1.6540	2,054.00 / 1.6548	1.00 / 0.81
MUX 2 / IN2	A	2,011.00 / 1.6202	2,025.00 / 1.6315	2,029.00 / 1.6347	4.00 / 3.22
MUX 2 / IN2	B	2,012.00 / 1.6210	2,022.00 / 1.6291	2,023.00 / 1.6299	1.00 / 0.81
MUX 3 / IN3	A	2,029.00 / 1.6347	2,033.00 / 1.6379	2,037.00 / 1.6411	4.00 / 3.22
MUX 3 / IN3	B	2,030.00 / 1.6355	2,028.00 / 1.6339	2,030.00 / 1.6355	2.00 / 1.61
MUX 4 / IN4	A	2,044.00 / 1.6468	2,051.00 / 1.6524	2,054.00 / 1.6548	3.00 / 2.42
MUX 4 / IN4	B	2,044.00 / 1.6468	2,054.00 / 1.6548	2,055.00 / 1.6556	1.00 / 0.81

A difference across ten conversion positions includes the transient response and conversion activity. Calling its SD a noise level would conflate recovery with noise. Every individual conversion and every repetition is retained in ten_conversion_repetitions.csv.

22. Array B - before baseline and burst endpoints

All targets are external MUX address 2. “First ADC” identifies the order of the two ten-reading bursts, not MUX number. The before baseline is from that same repetition’s continuous scan; reported values aggregate three repetitions.

MUX / ADC input	First ADC	Before baseline counts / V	First code counts / V	Tenth code counts / V	Tenth - first counts / mV
MUX 1 / IN1	A	1,924.00 / 1.5501	210.00 / 0.1692	276.00 / 0.2224	66.00 / 53.17
MUX 1 / IN1	B	1,921.00 / 1.5477	126.00 / 0.1015	194.00 / 0.1563	68.00 / 54.79
MUX 2 / IN2	A	1,988.00 / 1.6017	360.00 / 0.2900	489.00 / 0.3940	129.00 / 103.93
MUX 2 / IN2	B	1,985.00 / 1.5992	183.00 / 0.1474	327.00 / 0.2635	144.00 / 116.02
MUX 3 / IN3	A	1,928.00 / 1.5533	221.00 / 0.1781	288.00 / 0.2320	67.00 / 53.98
MUX 3 / IN3	B	1,924.00 / 1.5501	132.00 / 0.1063	203.00 / 0.1635	71.00 / 57.20
MUX 4 / IN4	A	1,993.00 / 1.6057	648.00 / 0.5221	828.00 / 0.6671	180.00 / 145.02
MUX 4 / IN4	B	1,987.00 / 1.6009	355.00 / 0.2860	596.00 / 0.4802	241.00 / 194.17

A difference across ten conversion positions includes the transient response and conversion activity. Calling its SD a noise level would conflate recovery with noise. Every individual conversion and every repetition is retained in ten_conversion_repetitions.csv.

23. Regular full cycles with 1, 5 and 10 ms VMID settling

Purpose

Test whether long VMID intervals between external MUX addresses change baseline, recorded temporal variation and full-cycle sampling rate.

What was done

1. Run a continuous full two-array scan. At each external MUX address 0-5, wait 5 µs and read all four ADC inputs on ADC A then ADC B; at address 6 read only IN4.

2. After every address, select external MUX address 7 on both arrays together. Wait the configured VMID delay, then read/discard IN1-IN4 on each ADC.

3. Repeat with VMID settling set separately to 1 ms, 5 ms and 10 ms. Keep sensor settling fixed at 5 µs and retain all 50 sensor readings per cycle.

Test parameters

Parameter	Setting
SPI / ordering	LPSPI; requested 16 MHz; both arrays; coordinated external-address switching; ADC A first; IN1-IN4; one sensor conversion per route.
Intervals per cycle	Seven sensor-address waits of 5 µs and seven shared VMID waits. VMID settles once per address for both arrays, not once per ADC input.
Recordings	Three 12 s recordings per VMID delay. Exclude the first 2 s; analyze the following 10 s. No fixed phase slots or cycle throttle.

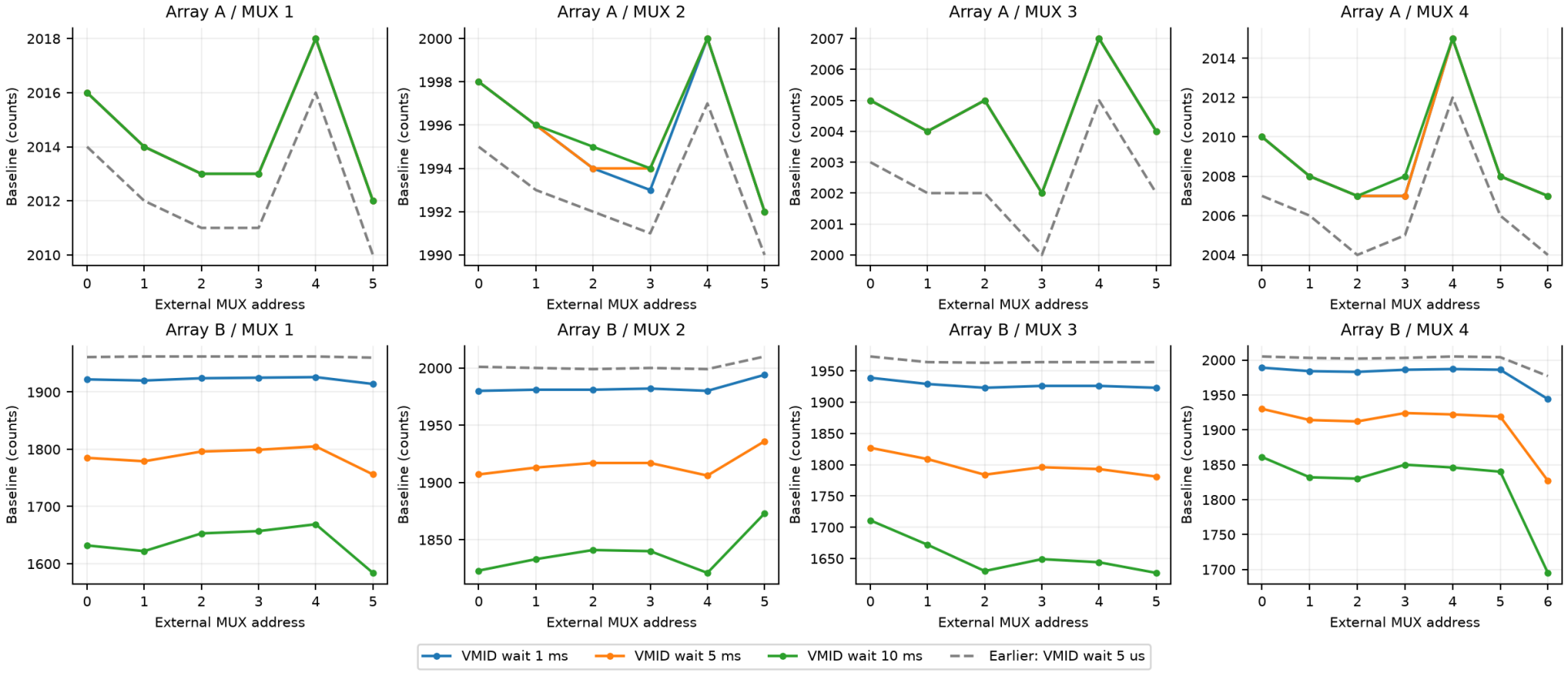
What was measured

Per-route median code, raw temporal SD and peak-to-peak in the analyzed window; spatial baseline spread across addresses; measured start-to-start cycle period. Drift is the last-two-second median minus the first-two-second median within that 10 s window. The initial exclusion does not establish equilibrium, and SD may include residual drift or isolated events.

How to read the results

The address graphs compare the same physical sensor baselines for each VMID delay. The center/noise graph uses external addresses 0-5 for equal six-sensor MUX summaries. The earlier 5 µs VMID curve is context from a separate recording session and a shorter analysis window, not a simultaneous matched-duration control.

23. Per-sensor baselines with long VMID settling - results



Legend and axes

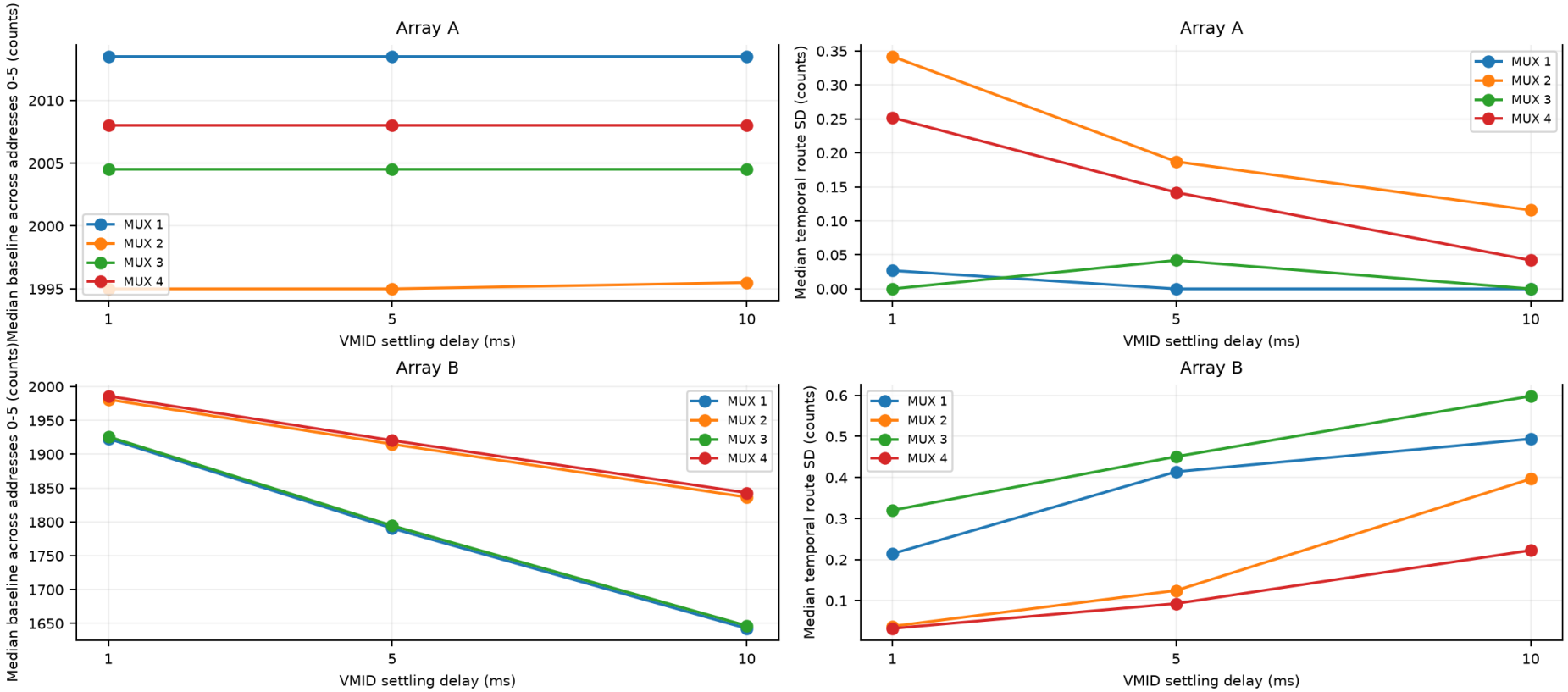
Colour identifies VMID wait 1, 5 or 10 ms. Grey dashed = earlier full scan with VMID wait 5 µs. X = external MUX address; Y = median sensor code. Lines connect different physical sensors.

Conclusions from these results

Changing VMID wait from 1 to 10 ms changes MUX baseline centers by +0.0 to +0.5 counts on Array A, and -280.5 to -143.0 counts on Array B. Longer dwell changes both reference-connection history and sensor disconnected time; these effects are not separated.

MUX 4's address-6 sensor remains included. Address 6 on MUXes 1-3 is VMID, so those reference responses are not substituted for sensor data.

23. MUX baseline centers and recorded temporal SD - results



Legend and axes

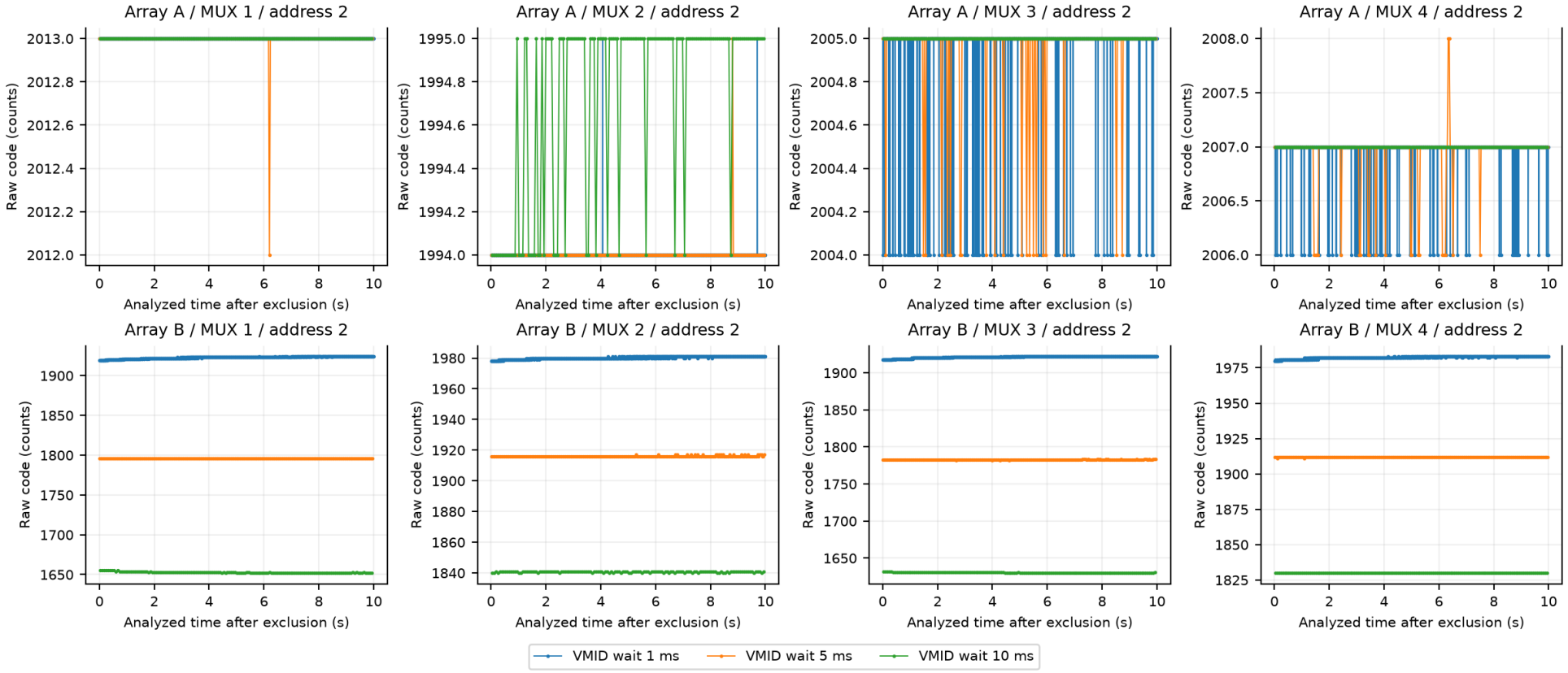
Each colour = MUX number. X = VMID settling delay in ms. Left = median baseline across six external addresses 0-5. Right = median of the six individual routes' temporal SDs, not SD across their baseline values.

Conclusions from these results

These are medians over a defined 10-second window, not a proof that every sensor reached a final equilibrium. Raw SD retains all excursions and any drift; it should not be called an isolated white-noise floor.

Longer VMID delay lowers the per-sensor sampling rate and observed bandwidth. A lower SD at a slower rate would not by itself demonstrate a quieter analog circuit.

23. Time traces: check for drift before calling variation noise - results



Legend and axes

Each dot is a sampled ADC code; colour = VMID wait 1, 5 or 10 ms. X = time within the analyzed 10-second window after the 2-second exclusion. The three curves come from separate recordings, not simultaneous measurements. Lines connect samples for visibility.

Conclusions from these results

Across addresses 0-5 and all repetitions, the largest absolute first-to-last two-second median change is 4 counts. That check separates a slow level change from a single SD summary; it is not a proof that every remaining fluctuation is white noise.

The examples show external MUX address 2 in recording 1 only. All routes and all three recordings contribute to the summary tables.

23. Sampling rate and configured wait contribution

VMID wait	Shared VMID wait / cycle µs	Sensor wait / cycle µs	Measured cycle µs	Cycles/s = samples/s per sensor	Retained sensor codes/s
1 ms	7000	35	7,191.00	139.063	6,953.1
5 ms	35000	35	35,191.00	28.416	1,420.8
10 ms	70000	35	70,193.00	14.246	712.3

Fit to these three measured periods: **cycle time approximately 7.00023 × VMID_wait_us + 190.44 µs**. The expected wait contribution is seven times the VMID setting because there are seven address groups per full cycle. The fitted remainder includes the 35 µs configured sensor waits, switching guards, SPI conversions, driver work and diagnostic transport overhead; it is not a direct measurement of each component.

Each cycle retains 50 sensor codes and performs another 56 discarded VMID conversions, for 106 conversions total. Cycle rate is therefore the sampling rate of each sensor, while 50 × cycle rate is the combined retained sensor-code rate. These are diagnostic-profile rates, not production-driver optimization measurements.

23. Baseline and variation summaries - addresses 0-5

Array / MUX	VMID wait ms	Baseline center counts / V	Spatial spread counts / mV	Median route SD counts / mV	Worst range counts / mV	Largest absolute drift counts	Samples / route / record
A / 1	1	2,013.50 / 1.6222	6.00 / 4.83	0.0268 / 0.0216	1 / 0.806	0.00	1,390
A / 1	5	2,013.50 / 1.6222	6.00 / 4.83	0.0000 / 0.0000	1 / 0.806	0.00	284
A / 1	10	2,013.50 / 1.6222	6.00 / 4.83	0.0000 / 0.0000	1 / 0.806	0.00	142
A / 2	1	1,995.00 / 1.6073	8.00 / 6.45	0.3419 / 0.2755	1 / 0.806	1.00	1,390
A / 2	5	1,995.00 / 1.6073	8.00 / 6.45	0.1872 / 0.1508	1 / 0.806	0.00	284
A / 2	10	1,995.50 / 1.6077	8.00 / 6.45	0.1157 / 0.0932	1 / 0.806	1.00	142
A / 3	1	2,004.50 / 1.6150	5.00 / 4.03	0.0000 / 0.0000	1 / 0.806	0.00	1,390
A / 3	5	2,004.50 / 1.6150	5.00 / 4.03	0.0419 / 0.0337	1 / 0.806	0.00	284
A / 3	10	2,004.50 / 1.6150	5.00 / 4.03	0.0000 / 0.0000	1 / 0.806	0.00	142
A / 4	1	2,008.00 / 1.6178	8.00 / 6.45	0.2519 / 0.2029	1 / 0.806	0.00	1,390
A / 4	5	2,008.00 / 1.6178	8.00 / 6.45	0.1419 / 0.1143	2 / 1.611	0.00	284
A / 4	10	2,008.00 / 1.6178	8.00 / 6.45	0.0420 / 0.0338	2 / 1.611	0.00	142
B / 1	1	1,923.00 / 1.5493	12.00 / 9.67	0.2142 / 0.1726	5 / 4.028	3.00	1,390
B / 1	5	1,790.50 / 1.4425	49.00 / 39.48	0.4139 / 0.3335	6 / 4.834	3.00	284
B / 1	10	1,642.50 / 1.3233	85.00 / 68.48	0.4940 / 0.3980	10 / 8.057	4.00	142
B / 2	1	1,981.00 / 1.5960	14.00 / 11.28	0.0378 / 0.0305	3 / 2.417	2.00	1,390
B / 2	5	1,915.00 / 1.5428	30.00 / 24.17	0.1249 / 0.1006	1 / 0.806	1.00	284
B / 2	10	1,836.50 / 1.4796	52.00 / 41.89	0.3963 / 0.3193	5 / 4.028	1.00	142
B / 3	1	1,926.00 / 1.5517	16.00 / 12.89	0.3201 / 0.2579	4 / 3.223	3.00	1,390
B / 3	5	1,794.50 / 1.4458	46.00 / 37.06	0.4508 / 0.3632	5 / 4.028	3.00	284
B / 3	10	1,646.50 / 1.3265	84.00 / 67.68	0.5981 / 0.4819	11 / 8.862	4.00	142
B / 4	1	1,986.00 / 1.6000	6.00 / 4.83	0.0324 / 0.0261	3 / 2.417	2.00	1,390
B / 4	5	1,920.50 / 1.5473	18.00 / 14.50	0.0931 / 0.0750	2 / 1.611	1.00	284
B / 4	10	1,843.00 / 1.4848	31.00 / 24.98	0.2226 / 0.1793	3 / 2.417	1.00	142

Temporal SD/range are computed for each sensor route within each recording, then aggregated over repetitions. “Median route SD” summarizes six routes; “spatial spread” compares their baseline values. Worst range and largest absolute drift select the largest value across all six routes and all three recordings. All 50 routes, including MUX 4 address 6, have counts and volts in long_scan_routes.csv and long_scan_repetitions.csv.

On Array B, the 1 Mohm paths (MUXes 1 and 3) fall by about 280 counts between 1 and 10 ms; the 470 kohm paths (MUXes 2 and 4) fall by about 144 counts. This pattern is consistent with an influence of bias strength and disconnected time, but does not prove that resistor value alone causes the differences.

VMID extension - acquisition evidence and limitations

Check / evidence	Result
Completed recordings	63: 27 continuous before scans, 27 single post-hold captures and 9 long-VMID regular scans. Smoke test excluded.
Hold check	27 timed holds; 20.000000 to 20.000001 s. Total recorded pre-to-post gaps: 21.675-21.799 s.
Acquired / received / discarded cycles	846,272 / 846,029 / 243
Transfer / ADC data / USB-write errors / timing overruns	0 / 0 / 0 / 0
Single-frame verification	Every post-hold recording has exactly one complete transmitted cycle, sequence zero; no dropped post-hold cycles.
Firmware isolation	pcbdiag-v2 adds single-capture/long-hold control and ten repeated responses. Hold occurs outside per-cycle CPU-tick timing; selectors are prepared before the hold. Production protocol unchanged.
Validation	84 focused regression tests passed, including the actual C++ diagnostic model with a simulated 20 s hold and ten-response bursts. Production and diagnostic builds passed.
Restoration	production_restored_verified; adc124-selector-lookahead-v1; acquisition stopped
Data folder	docs/testing/datasets/testboard-adc124/vmid_long_hold_2026-10-15
Results folder	docs/testing/results/testboard-adc124/vmid_long_hold_2026-10-15

The analyzed before-window defines a reproducible scan history, but the 20-second reference interval is not an isolated charge-injection test: sensors are disconnected throughout parking. Host setup adds parked time before the exact timed hold. No sensor was intentionally pressed; mechanical disturbances were not instrumented. The arrays' different MUX types remain confounded with their complete analog paths and PCB locations.

Original matrix: acquisition checks, restoration and evidence files

Purpose: confirm that the plotted codes belong to fully decoded recordings, and that the temporary experiment did not leave the board running diagnostic firmware.

Check	Result
Expected / complete recordings	762 / 762
Acquired / received / discarded cycles	7,919,928 / 7,919,811 / 117; every recording accounts for all acquired cycles
Transfer / data / USB-write errors	0 / 0 / 0
Timing overruns	0
Frame validation	ADG1 schema, per-conversion labels, monotonic sequence and FNV-1a checksum checked
Firmware checks	82 focused regression tests passed; tagged diagnostic model checked 810 configurations × three cycles; production and diagnostic builds passed
Restored state	production_restored_verified; adc124-selector-lookahead-v1; stopped
Original settings	Restored, including original engine/scan order; not forced to the diagnostic LPSPI settings

One separate smoke test is excluded. An incomplete long-isolation attempt hit a host timeout; its bytes are preserved separately, the reader was corrected, production was restored, and the condition was repeated. Only complete recordings are included.

Conclusion: no recorded framing fault or controlled timing overrun explains the observed ADC-code events. Quiet sensor data alone do not independently prove analog channel identity; physical labels rely on the documented wiring and validated digital sequencing.

How to find the underlying data

File / location	Contents
docs/testing/results/testboard-adc124/pcb_diagnostics_2026-10-15	This report’s analysis, tables, figures, reader explanations and verification record.
channels.csv / channel_repetitions.csv	Per array, MUX number, external MUX address and repeated conversion: baseline, SD, range, counts/volts, repetition variation and timing markers.
captures.csv / analysis.json	All conditions, programmed steps, measured cycle periods and acquisition accounting.
long_noise.csv / long_spikes.csv	SD-selected examples versus independently peak-to-peak-selected examples.
long_quiet_channels.csv / long_quiet_metadata.json	Every long-record route’s quiet statistics and exact screening intervals/thresholds.
correlations.csv / timing_correlations.csv	Pairwise and code/timing associations used in Tests 19 and 20.
docs/testing/datasets/testboard-adc124/pcb_diagnostics_2026-10-15	Original raw bytes, decoded arrays/labels, capture metadata, command/upload logs, production backup and diagnostic firmware/tool snapshots.
report_revisions/original	The earlier PDF, builder and verification record, preserved before this explanatory rewrite.

Tests 1-20 retain the original recordings. New Tests 21-23 have separate raw/analysis folders listed in their acquisition-evidence page. The temporary diagnostic image was updated for those tests, then original production firmware/settings were restored.

Device references and limits of component attribution

Purpose: distinguish the observed board behaviour from claims that would need a component-level measurement. The following manufacturer documents are background references, not measurements of this assembled board.

ADC124S101 data sheet - https://www.ti.com/lit/ds/symlink/adc124s101.pdf

TMUX1108 data sheet - https://www.ti.com/lit/ds/symlink/tmux1108.pdf

TMUX1308A data sheet - https://www.ti.com/lit/ds/symlink/tmux1308a.pdf

OPA365 data sheet - https://www.ti.com/lit/ds/symlink/opa365.pdf

REF2033 family data sheet - https://www.ti.com/lit/pdf/sbos600

High-impedance MUX settling application note - https://www.ti.com/lit/pdf/SCDA054

ADC124S101’s command selects the next conversion input. Its data sheet does not require a power-up dummy conversion. A measured improvement after more conversions is therefore not evidence of a digital requirement to discard one reading. The OPA365 buffer also separates the external high-resistance MUX node from the ADC sampling capacitor.

Conclusion: these automatic controls establish sensitivity to operating history. They do not separately measure charge injection, PZT capacitance, leakage, reference ripple, buffer stability or physical GPIO/SPI waveforms. Controlled component substitution or probing is needed before assigning one component as the cause.

Appendix: original programmed conditions and measured cycle periods

This appendix is an audit index, not a graph legend. Program names are machine identifiers linked to raw recording filenames. Arrays A, B or A+B identify enabled arrays. A requested period of “free” means no programmed throttle; measured values include diagnostic overhead. Clock values are requested MHz. Phase count includes read and no-read steps.

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
boundary_address6_1	boundary_matched	3	16	A+B	7	710	710.00
boundary_address6_50	boundary_matched	3	16	A+B	7	710	710.00
boundary_address6_5	boundary_matched	3	16	A+B	7	710	710.00
boundary_disabled_1	boundary_matched	3	16	A+B	7	710	710.00
boundary_disabled_50	boundary_matched	3	16	A+B	7	710	710.00
boundary_disabled_5	boundary_matched	3	16	A+B	7	710	710.00
boundary_hold_1	boundary_matched	3	16	A+B	7	710	710.00
boundary_hold_50	boundary_matched	3	16	A+B	7	710	710.00
boundary_hold_5	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_1	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_50	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_5	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_adc_1	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_adc_50	boundary_matched	3	16	A+B	7	710	710.00
boundary_vmid_adc_5	boundary_matched	3	16	A+B	7	710	710.00
fast_core_1	boundary_fast	3	16	A+B	6	free	78.00
fast_core_50	boundary_fast	3	16	A+B	6	free	372.00
fast_core_5	boundary_fast	3	16	A+B	6	free	102.00
fast_full_1	boundary_fast	3	16	A+B	7	free	85.00
fast_full_50	boundary_fast	3	16	A+B	7	free	428.00
fast_full_5	boundary_fast	3	16	A+B	7	free	113.00
fast_full_vmid_1	boundary_fast	3	16	A+B	14	free	171.00
fast_full_vmid_50	boundary_fast	3	16	A+B	14	free	856.00
fast_full_vmid_5	boundary_fast	3	16	A+B	14	free	226.00
fast_order_core_gray	fast_order	3	16	A+B	6	free	102.00
fast_order_full_descending	fast_order	3	16	A+B	7	free	113.00
fast_order_full_rotated	fast_order	3	16	A+B	7	free	113.00
held_a0_mask15	held	3	16	A+B	1	710	710.00
held_a0_mask1	held	3	16	A+B	1	710	710.00
held_a0_mask2	held	3	16	A+B	1	710	710.00
held_a0_mask4	held	3	16	A+B	1	710	710.00
held_a0_mask8	held	3	16	A+B	1	710	710.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
held_a2_mask15	held	3	16	A+B	1	710	710.00
held_a2_mask1	held	3	16	A+B	1	710	710.00
held_a2_mask2	held	3	16	A+B	1	710	710.00
held_a2_mask4	held	3	16	A+B	1	710	710.00
held_a2_mask8	held	3	16	A+B	1	710	710.00
held_a4_mask15	held_extra	3	16	A+B	1	710	710.00
held_a5_mask15	held	3	16	A+B	1	710	710.00
held_a5_mask1	held	3	16	A+B	1	710	710.00
held_a5_mask2	held	3	16	A+B	1	710	710.00
held_a5_mask4	held	3	16	A+B	1	710	710.00
held_a5_mask8	held	3	16	A+B	1	710	710.00
known_held6_1_banks3	known_vmid	3	16	A+B	1	710	710.00
known_held6_50_banks3	known_vmid	3	16	A+B	1	710	710.00
known_held6_5_banks3	known_vmid	3	16	A+B	1	710	710.00
known_held7_1_banks3	known_vmid	3	16	A+B	1	710	710.00
known_held7_50_banks3	known_vmid	3	16	A+B	1	710	710.00
known_held7_5_banks3	known_vmid	3	16	A+B	1	710	710.00
known_toggle67_1_banks3	known_vmid	3	16	A+B	2	710	710.00
known_toggle67_50_banks3	known_vmid	3	16	A+B	2	710	710.00
known_toggle67_5_banks1	known_vmid	3	16	A	2	710	710.00
known_toggle67_5_banks2	known_vmid	3	16	B	2	710	710.00
known_toggle67_5_banks3	known_vmid	3	16	A+B	2	710	710.00
leak_b0_m0_1000000	leakage	3	16	A	3	1100500	1,100,500.00
leak_b0_m0_100000	leakage	3	16	A	3	200500	200,500.00
leak_b0_m0_10000	leakage	3	16	A	3	110500	110,500.00
leak_b0_m0_1000	leakage	3	16	A	3	101500	101,500.00
leak_b0_m1_1000000	leakage	3	16	A	3	1100500	1,100,500.00
leak_b0_m1_100000	leakage	3	16	A	3	200500	200,500.00
leak_b0_m1_10000	leakage	3	16	A	3	110500	110,500.00
leak_b0_m1_1000	leakage	3	16	A	3	101500	101,500.00
leak_b0_m2_1000000	leakage	3	16	A	3	1100500	1,100,500.00
leak_b0_m2_100000	leakage	3	16	A	3	200500	200,500.00
leak_b0_m2_10000	leakage	3	16	A	3	110500	110,500.00
leak_b0_m2_1000	leakage	3	16	A	3	101500	101,500.00
leak_b0_m3_1000000	leakage	3	16	A	3	1100500	1,100,500.00
leak_b0_m3_100000	leakage	3	16	A	3	200500	200,500.00
leak_b0_m3_10000	leakage	3	16	A	3	110500	110,500.00
leak_b0_m3_1000	leakage	3	16	A	3	101500	101,500.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
leak_b1_m0_1000000	leakage	3	16	B	3	1100500	1,100,500.00
leak_b1_m0_100000	leakage	3	16	B	3	200500	200,500.00
leak_b1_m0_10000	leakage	3	16	B	3	110500	110,500.00
leak_b1_m0_1000	leakage	3	16	B	3	101500	101,500.00
leak_b1_m1_1000000	leakage	3	16	B	3	1100500	1,100,500.00
leak_b1_m1_100000	leakage	3	16	B	3	200500	200,500.00
leak_b1_m1_10000	leakage	3	16	B	3	110500	110,500.00
leak_b1_m1_1000	leakage	3	16	B	3	101500	101,500.00
leak_b1_m2_1000000	leakage	3	16	B	3	1100500	1,100,500.00
leak_b1_m2_100000	leakage	3	16	B	3	200500	200,500.00
leak_b1_m2_10000	leakage	3	16	B	3	110500	110,500.00
leak_b1_m2_1000	leakage	3	16	B	3	101500	101,500.00
leak_b1_m3_1000000	leakage	3	16	B	3	1100500	1,100,500.00
leak_b1_m3_100000	leakage	3	16	B	3	200500	200,500.00
leak_b1_m3_10000	leakage	3	16	B	3	110500	110,500.00
leak_b1_m3_1000	leakage	3	16	B	3	101500	101,500.00
long_held	long_noise	1	16	A+B	1	1000	1,000.00
long_scan	long_noise	1	16	A+B	6	1000	1,000.00
long_scan_vmid	long_noise	1	16	A+B	12	1000	1,000.00
order_ascending_ref0	order	3	16	A+B	7	710	710.00
order_ascending_ref1	order	3	16	A+B	7	710	710.00
order_descending_ref0	order	3	16	A+B	7	710	710.00
order_descending_ref1	order	3	16	A+B	7	710	710.00
order_gray_ref0	order	3	16	A+B	7	710	710.00
order_gray_ref1	order	3	16	A+B	7	710	710.00
order_permuted_ref0	order	3	16	A+B	7	710	710.00
order_permuted_ref1	order	3	16	A+B	7	710	710.00
order_rotated_ref0	order	3	16	A+B	7	710	710.00
order_rotated_ref1	order	3	16	A+B	7	710	710.00
readorder_1_banks1_first0_order228	readorder	3	16	A	7	710	710.00
readorder_1_banks2_first0_order228	readorder	3	16	B	7	710	710.00
readorder_1_banks3_first0_order228	readorder	3	16	A+B	7	710	710.00
readorder_1_banks3_first0_order27	readorder	3	16	A+B	7	710	710.00
readorder_1_banks3_first1_order228	readorder	3	16	A+B	7	710	710.00
readorder_50_banks1_first0_order228	readorder	3	16	A	7	710	710.00
readorder_50_banks2_first0_order228	readorder	3	16	B	7	710	710.00
readorder_50_banks3_first0_order228	readorder	3	16	A+B	7	710	710.00
readorder_50_banks3_first0_order27	readorder	3	16	A+B	7	710	710.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
readorder_50_banks3_first1_order228	readorder	3	16	A+B	7	710	710.00
readorder_5_banks1_first0_order228	readorder	3	16	A	7	710	710.00
readorder_5_banks2_first0_order228	readorder	3	16	B	7	710	710.00
readorder_5_banks3_first0_order228	readorder	3	16	A+B	7	710	710.00
readorder_5_banks3_first0_order27	readorder	3	16	A+B	7	710	710.00
readorder_5_banks3_first1_order228	readorder	3	16	A+B	7	710	710.00
recovery_b0_m0_16000000_1_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m0_16000000_1_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m0_16000000_1_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m0_16000000_50_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m0_16000000_50_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m0_16000000_50_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m0_16000000_5_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m0_16000000_5_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m0_16000000_5_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m0_8000000_1_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m0_8000000_1_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m0_8000000_1_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m0_8000000_5_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m0_8000000_5_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m0_8000000_5_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m1_16000000_1_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m1_16000000_1_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m1_16000000_1_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m1_16000000_50_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m1_16000000_50_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m1_16000000_50_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m1_16000000_5_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m1_16000000_5_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m1_16000000_5_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m1_8000000_1_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m1_8000000_1_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m1_8000000_1_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m1_8000000_5_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m1_8000000_5_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m1_8000000_5_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m2_16000000_1_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m2_16000000_1_wait_plus	recovery_bracket	3	16	A	2	250	250.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
recovery_b0_m2_16000000_1_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m2_16000000_50_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m2_16000000_50_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m2_16000000_50_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m2_16000000_5_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m2_16000000_5_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m2_16000000_5_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m2_8000000_1_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m2_8000000_1_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m2_8000000_1_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m2_8000000_5_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m2_8000000_5_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m2_8000000_5_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m3_16000000_1_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m3_16000000_1_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m3_16000000_1_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m3_16000000_50_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m3_16000000_50_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m3_16000000_50_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m3_16000000_5_eight	recovery	3	16	A	2	250	250.00
recovery_b0_m3_16000000_5_wait_plus	recovery_bracket	3	16	A	2	250	250.00
recovery_b0_m3_16000000_5_wait	recovery	3	16	A	2	250	250.00
recovery_b0_m3_8000000_1_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m3_8000000_1_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m3_8000000_1_wait	recovery	3	8	A	2	250	250.00
recovery_b0_m3_8000000_5_eight	recovery	3	8	A	2	250	250.00
recovery_b0_m3_8000000_5_wait_plus	recovery_bracket	3	8	A	2	250	250.00
recovery_b0_m3_8000000_5_wait	recovery	3	8	A	2	250	250.00
recovery_b1_m0_16000000_1_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m0_16000000_1_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m0_16000000_1_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m0_16000000_50_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m0_16000000_50_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m0_16000000_50_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m0_16000000_5_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m0_16000000_5_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m0_16000000_5_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m0_8000000_1_eight	recovery	3	8	B	2	250	250.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
recovery_b1_m0_8000000_1_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m0_8000000_1_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m0_8000000_5_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m0_8000000_5_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m0_8000000_5_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m1_16000000_1_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m1_16000000_1_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m1_16000000_1_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m1_16000000_50_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m1_16000000_50_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m1_16000000_50_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m1_16000000_5_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m1_16000000_5_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m1_16000000_5_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m1_8000000_1_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m1_8000000_1_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m1_8000000_1_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m1_8000000_5_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m1_8000000_5_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m1_8000000_5_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m2_16000000_1_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m2_16000000_1_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m2_16000000_1_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m2_16000000_50_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m2_16000000_50_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m2_16000000_50_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m2_16000000_5_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m2_16000000_5_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m2_16000000_5_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m2_8000000_1_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m2_8000000_1_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m2_8000000_1_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m2_8000000_5_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m2_8000000_5_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m2_8000000_5_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m3_16000000_1_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m3_16000000_1_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m3_16000000_1_wait	recovery	3	16	B	2	250	250.00

Program identifier	Family	Reps	MHz	Arrays	Phases	Requested period µs	Measured median period µs
recovery_b1_m3_16000000_50_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m3_16000000_50_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m3_16000000_50_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m3_16000000_5_eight	recovery	3	16	B	2	250	250.00
recovery_b1_m3_16000000_5_wait_plus	recovery_bracket	3	16	B	2	250	250.00
recovery_b1_m3_16000000_5_wait	recovery	3	16	B	2	250	250.00
recovery_b1_m3_8000000_1_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m3_8000000_1_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m3_8000000_1_wait	recovery	3	8	B	2	250	250.00
recovery_b1_m3_8000000_5_eight	recovery	3	8	B	2	250	250.00
recovery_b1_m3_8000000_5_wait_plus	recovery_bracket	3	8	B	2	250	250.00
recovery_b1_m3_8000000_5_wait	recovery	3	8	B	2	250	250.00
refburst_banks1_first0_n1	refburst	3	16	A	2	250	250.00
refburst_banks1_first0_n8	refburst	3	16	A	2	250	250.00
refburst_banks2_first0_n1	refburst	3	16	B	2	250	250.00
refburst_banks2_first0_n8	refburst	3	16	B	2	250	250.00
refburst_banks3_first0_n1	refburst	3	16	A+B	2	250	250.00
refburst_banks3_first0_n8	refburst	3	16	A+B	2	250	250.00
refburst_banks3_first1_n8	refburst	3	16	A+B	2	250	250.00
reset_control_idle_1	reset_control	3	16	A+B	12	920	920.00
reset_control_idle_50	reset_control	3	16	A+B	12	920	920.00
reset_control_idle_5	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_1	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_50	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_5	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_adc_1	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_adc_50	reset_control	3	16	A+B	12	920	920.00
reset_control_vmid_adc_5	reset_control	3	16	A+B	12	920	920.00
reset_short_idle_1	reset_short	3	16	A+B	12	300	300.00
reset_short_idle_5	reset_short	3	16	A+B	12	300	300.00
reset_short_vmid_1	reset_short	3	16	A+B	12	300	300.00
reset_short_vmid_5	reset_short	3	16	A+B	12	300	300.00
reset_short_vmid_adc_1	reset_short	3	16	A+B	12	300	300.00
reset_short_vmid_adc_5	reset_short	3	16	A+B	12	300	300.00